

Nephrology

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Executive or Committee Positions



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Clinical Expert

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Advisory Board*



Travel Support

Overview

1. Acid Base
 2. Hypo/Hybernatriemia
 3. Hypo/Hyperkalemia
 4. Hypertension
 5. Glomerular Disease
 6. AKI
 7. CKD Progression
 8. CKD Complications
- BONUS Nephro Nuggets
 - Acronyms used
 - “Lumps” in the kidney
 - ADPKD
 - Choosing Wisely
 - Nephrolithiasis
 - HTN (practical)
 - When to biopsy
 - Urine lytes/Hypokalemia (bonus)
 - Hematologic malignancies and the kidney
 - Landmark trials
 - Bonus MCQ Cases

BONUS

Read on own

Acronyms Used in this Talk

AOBP	Automated Office Blood Pressure
ABPM	Ambulatory Blood Pressure Monitor (usually 24 hr)
AG (UAG)	Anion Gap (Urinary Anion Gap)
AIN	Acute Interstitial nephritis
CHEP	Canadian Hypertension Education Program
ESA	Erythropoietin Stimulating Agent
EGPA	Eosinophilic Granulomatosis w/ Polyangiitis (Churg-Straus)
FSGS	Focal Segmental Glomerulosclerosis
GPA	Granulomatosis with Polyangiitis (Wegner's)
HTN	Hypertension
HBPM	Home Blood Pressure Measurement
MBD	Metabolic Bone Diseases
MPA	Microscopic Polyangiitis
OBPM	Office Blood Pressure Measurement
RAS	Renal Artery Stenosis
RTA	Renal Tubular Acidosis

Guidelines reviewed



Hypertension – 2025
Primary care update



KDIGO (kdigo.org)

CKD 2024

GN –

- ANCA 2024
- Lupus 2024
- IgA Nephropathy – 2024
- GN – 2021 (Membranous, FSGS, MCD, MPGN)

Anemia

ADPKD - 2025

1. Acid/Bases



Approach to Nephro Labs

- Acid/Base Disorders
- Hyponatremia
- Hypernatremia
- Hyperkalemia
- Hypokalemia

There are several practice MCQs and scenarios in the bonus slides to cement the concepts herein!

Acid-base:



pH

<7.4

>7.4

PCO_2 is $\uparrow$
= Respiratory
acidosis

HCO_3 is $\downarrow$
= Metabolic
acidosis

HCO_3 is $\downarrow$
 PCO_2 is $\uparrow$
= MIXED

HCO_3 is $\uparrow$
 PCO_2 normal **or** $\uparrow$
= METABOLIC alkalosis

Metabolic Acidosis next steps:
i) Appropriate compensation?
ii) Anion Gap? (Albumin)
iii) delta-delta? $\Delta\text{AG}:\Delta\text{bicarb}$
iv) Osmolar gap?

PCO_2 is $\downarrow$
 HCO_3 normal or $\downarrow$
= RESPIRATORY alkalosis

Acid/Base Disorders

1. Determine the primary disorder:

- Look at the pH
- Then look at HCO_3^- (24) & pCO_2 (45)

$\Delta\text{pCO}_2 : \Delta\text{HCO}_3^-$

	RESPIRATORY ACIDOSIS	RESPIRATORY ALKALOSIS
ACUTE	10:1	10:2
CHRONIC	10:3-4	10:4-5
	For every rise of 10 of pCO_2 the HCO_3^- will rise by 1 (or 3-4) $\Delta\text{pCO}_2 : \Delta\text{HCO}_3^-$	For every fall of 10 of pCO_2 the HCO_3^- will fall by 2 (or 4-5) $\Delta\text{pCO}_2 : \Delta\text{HCO}_3^-$

2. Compensation?

- Metabolic Acidosis ($\downarrow \text{HCO}_3^- : \downarrow \text{CO}_2$) 1:~1
- Metabolic Alkalosis ($\uparrow \text{HCO}_3^- : \uparrow \text{CO}_2$) 1:0.7 (1:~1)

- Respiratory Alkalosis ($\downarrow \text{CO}_2 : \downarrow \text{HCO}_3^-$) Chronic (10:4-5) | Acute (10:2)
- Respiratory Acidosis ($\uparrow \text{CO}_2 : \uparrow \text{HCO}_3^-$) Chronic (10: 3-4) | Acute (10:1)

Metabolic Acidosis

1. Calculate the **anion gap**: $\text{Na} - \text{Cl} - \text{HCO}_3$
 - Adjust for albumin (every $\downarrow$ in albumin by 10, add 2.5mEq/L to the AG)
2. Calculate the “**delta-delta**”: ΔAG (12-AG): Δbicarb (24-bicarb)
 - The **delta AG** (from normal = 12) should be equal to the change in **bicarb** (from normal = 24) when pure AGMA
 - See next slide for examples
3. Calculate the **osmol gap**: $[\text{calculated osm: } 2 \times \text{Na} + \text{Gluc} + \text{BUN}] - [\text{serum measured osm}]$
 - Normal osmolar gap = 10; higher means additional unmeasured osmoles– think EtOH or **toxic alcohol (dangerous to miss!)**
 - Could have high osmolar gap with normal anion gap early in toxic alcohol

Calculated osm= 2 salts and a sticky Bun!

Delta:Delta

$$\begin{aligned}\Delta/\Delta &= \Delta AG / \Delta HCO_3 \\ &= \frac{\text{measured AG} - \text{normal AG}}{\text{normal HCO}_3 - \text{measured HCO}_3} \\ &= \frac{[AG] - 12}{24 - [HCO_3]}\end{aligned}$$

Delta:Delta $\Delta AG : \Delta HCO_3$	Interpretation
$\Delta AG \gg \Delta HCO_3$	>2, bicarb doesn't change enough, meaning a secondary alkalosis is opposing the acidosis Concurrent Metabolic alkalosis (HCO_3 higher than expected) with anion gap metabolic acidosis
$\Delta AG \approx \Delta HCO_3$	0.8-2, Pure AG acidosis
$\Delta AG \ll \Delta HCO_3$	<0.8, Bicarb changes more than expected, meaning a secondary acidosis is present Concurrent non AG metabolic acidosis (with HCO_3 lower than expected) with high AG acidosis

EX: pH = **7.28**, **HCO₃ = 14** pCO₂ = 30
Na = 142, Cl = 95. Alb = 40. BUN 8. Glc 5
Serum Osm: 306

$$\Delta/\Delta = \frac{\Delta AG (33-12)}{\Delta HCO_3 (24-14)} = \frac{21}{10} = \text{Ratio} > 2$$

$$\Delta AG \gg \Delta HCO_3$$

Bicarb is higher than expected: something driving the HCO_3 up
= concomitant metabolic alkalosis

pH 7.28, $HCO_3 \downarrow$ = metab. acidosis
AG = 142-14-95 = 33
Osm Gap = 306 - ((142x2) + 8 + 5) = 9
Compensation 1:1
($\Delta HCO_3 = 24-14 = 10$: $\Delta CO_2 = 40-30 = 10$)

Causes of AGMA (GOLDMARK)

G- Glycols*

O- Oxoproline

L- Lactate

D- D-lactate

M- Methanol*

A- ASA

R- Renal failure

K- Ketones

Other mnemonics: MUDPILES,
KARMEL

***Ethylene Glycol & Methanol =
High Osmolar Gap**

Oxoproline – rare cause of AGMA

- organic acid
- byproduct of gamma-glutamyl cycle (process that transports amino acids across cell membranes)
- cycle produces glutathione

Consider if unexplained AGMA in setting of Chronic ACETAMINOPHEN USE

(even at 3-4g/d) can deplete glutathione = lose –ve feedback on cycle = excess Oxoproline (pyroglutamic acid)
= renal failure exacerbates this (unable to excrete oxoproline)

Other causes – pregnancy, vegan, malnutrition, deplete glycine (required for glutathione synthesis)

Causes of Increased Osmolar Gap

Anion Gap Metabolic Acidosis	No Metabolic Acidosis
Organic alcohol poisoning: • Methanol • Ethylene glycol Paraldehyde Ketoacidosis (EtOH ¹ + Diabetic) Lactic acidosis Severe CKD	Ethanol ¹ Isopropyl alcohol Mannitol Sorbitol *pseudohyponatremia *Early toxic alcohol! Hyperlipidemia Hyperproteinemia (ex myeloma)

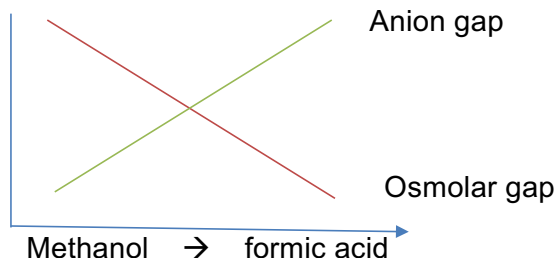
AGMA + OSM GAP = R/O TOXIC ALCOHOL

***More on this in Critical Care Lecture**

¹ EtOH can cause both AGMA & NAGMA

Acid/Base Disorders: Pearls

- In organic alcohol intoxications, the osmolar gap and anion gap may exist at different times
 - **Early on: osmolar gap without anion gap**
 - **Later: anion gap without osmolar gap**



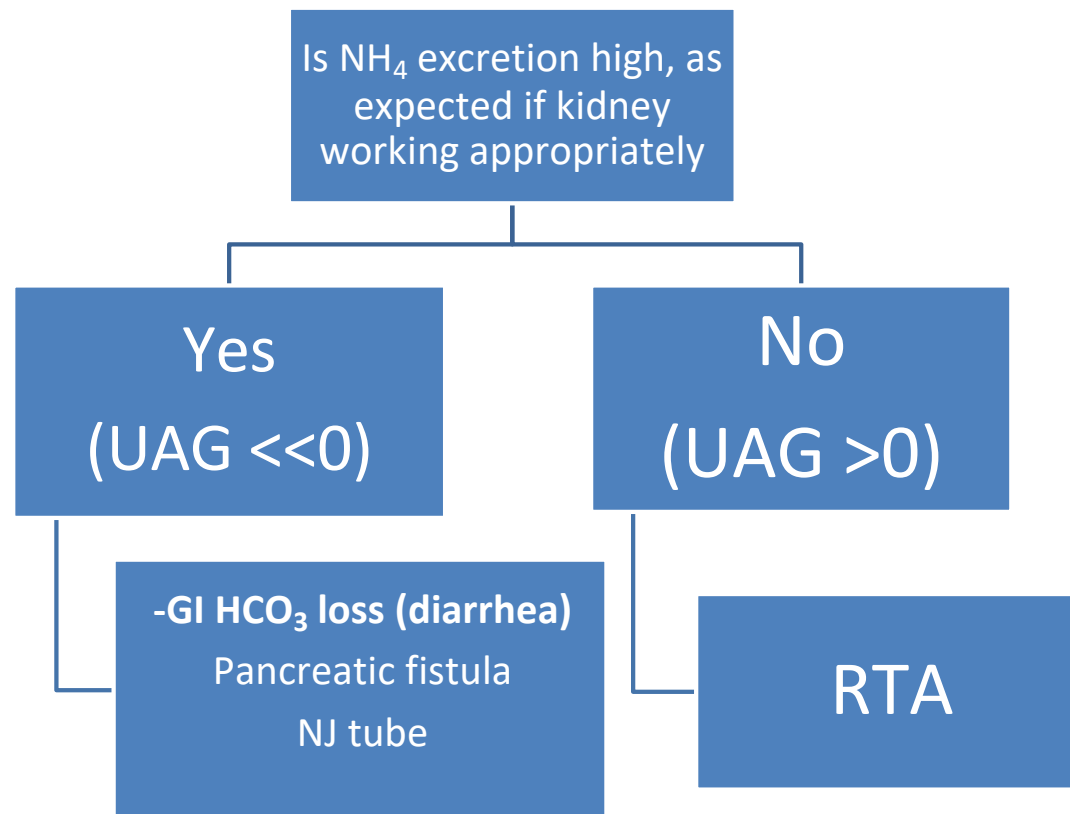
This is because the parent alcohol (ie- Methanol) is an OSMOLE and gets metabolized to the toxic ACID (formic acid)

- If you see an ABG with pH 7.4, be alert for mixed disorders
 - E.g. if the bicarb is low, go through all of the steps for metabolic acidosis

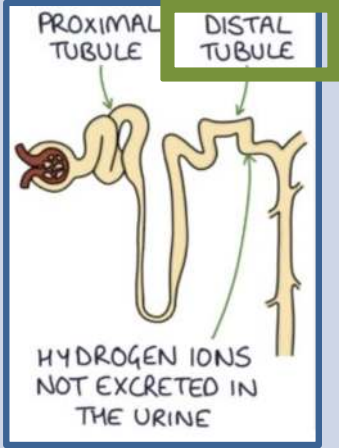
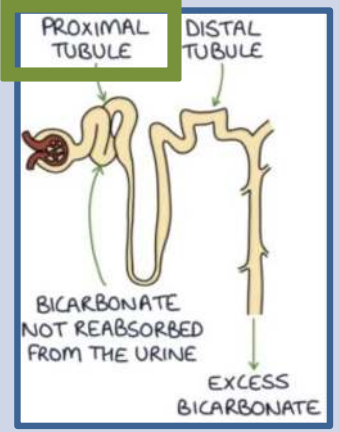
NAGMA Approach

Check urinary anion gap:
Urine(Na⁺) + Urine(K⁺) – Urine(Cl)

*Urine anion gap: If its Negative its
Not the kidney!
NEGATIVE = the gut*



RTA: Memorize Now; Forget in June

Lab	Type I	Type II	Type IV
Pathophysiology	DISTAL: ↓ in H ⁺ excretion 	PROXIMAL: ↓ in bicarb reabsorption 	Hypoaldosterone state impaired urinary ammonium excretion
HCO ₃ range	May be <10	12-20	~ >17
K range (serum)	Severely low K	Low	High or high normal
Urine pH	>5.5	Low , can still acidify urine	Variable
Other findings	Calcium phosphate stones	Low serum PO ₄ , low vit D, glucosuria	
Causes	CTD (Sjogren's , RA, SLE) Hypercalciuria, Drugs (ifosfamide)	Fanconi's (children) Myeloma , Acetazolamide, Tenofovir	Drugs: MRA, RASi, heparin , CNI, adrenal insufficiency (AI), Diabetes
Treatment	NaHCO₃ K Citrate (if K low)	NaHCO₃ K Citrate	NaHCO₃ Tx of AI: Florinef

Metabolic Alkalosis

UCI>20 Low BP Memory Tool:
B is before G (i.e. Barter's before Gittleman's). Barter's acts on the Ascending Loop)
L is before T (i.e. Lasix before thiazide) and Lasix acts on the Ascending Loop)

**Metabolic Alkalosis
(often with
hypoKalemia)**

**Normal BP /
Hypotension**

Urine Chloride <20

Vomiting or intermittent diuretic use
(diuretic already "worn off")

Urine Chloride > 20

Barter, Gitelman or recent diuretic use

Hypertension

Low Renin/ Low Aldo: e.g. Cushings,
Liddles, Licorice, Florinef, steroids

Low Renin/ high Aldo:
Adrenal problem e.g. Conns

High Renin/ high Aldo:
e.g. RAS/Reninoma

Genetic Channelopathies (learn now, forget in June)

	Bartter Syndrome Autosomal recessive	Gitelman's Syndrome Autosomal recessive	Liddle's Autosomal dominant
Defect	"Like Lasix" – ascending loop of henle **aminoglycoside use → "acquired" Bartter's	Distal tubule Na/Cl cotransporter "Like TZD"	ENaC mutation in Collecting duct – reabsorb too much Na
Labs and BP All cause: Hypokalemic Metabolic alkalosis	↓ K, ↓ Cl ↓ Ca ↓ Mg *high calcium in urine* ↓ or normal BP ↑ Renin ↑ Aldo	↓ K ↓ Cl ↓ Mg *low calcium in urine* ↓ or normal BP ↑ Renin ↑ Aldo	↓ K ↑ BP ↓ Renin ↓ Aldo *HTN* *clinically acts like hyperaldosterone, but serum aldo is low*
Treatment	K, Mg replacement K sparing diuretics, RASi NSAIDS	K, Mg replacement K sparing diuretics, RASi NSAIDS	Amiloride

Bartter loops out salt,
Gitelman leaks salt late,
Liddle locks salt in — High BP is their fate.

2. Hypo/Hyponatremia



Hyponatremia

- The most important questions to answer on initial assessment:

- Is this a medical emergency?
- Is this symptomatic or asymptomatic?
- Is this acute, chronic, or acute-on-chronic?
 - Acute <48hrs

TBW= Total Body Water

Female- 50% of Wt (Kg) or $Wt (Kg) \times 0.5$

Male- 60% of Wt (Kg) or $Wt (Kg) \times 0.6$

- Hyponatremia correction formula:

- Volume infusate to give = $\frac{TBW \times (\text{desired Na} - \text{Serum Na})}{[\text{Na}] \text{ infusate}}$

- Infusate = solution you choose to give
 - Hypertonic saline 3%- 513mmol/L Na
 - Normal Saline 0.9% - 154mmol/L Na
 - Ringers Lactate- 130mmol/L Na

Hyponatremia

(Symptomatic vs asymptomatic)

If symptomatic/acute → 3% Hypertonic Saline → raise Serum Na by 4.

- If seizing in oral scenario (or real life):
 - “Bolus 100ml Hypertonic (3%) saline over 10 min” – repeat up to 2 times if symptoms persist (max 300 ml/30min)

If not symptomatic/acute → there is time →

- Send tests (Uosm, S osm, UNa)
- Stop thiazides
- Thorough history (medications, diet, fluid intake, volume status)

Hyponatremia: Severe Symptoms

Example calculation:

- Severe and symptomatic (usually also acute).
- (ie) 60 kg woman presenting with seizures, Na of 110.
 - Goal: increase serum Na by 4-5 mmol/L immediately
 - **Volume infusate to give = $\frac{\text{TBW} \times (\text{desired Na} - \text{Serum Na})}{[\text{Na}] \text{ infusate}}$**
 - = $\frac{(60\text{kg} \times 0.5) \times (115 - 110)}{513}$
 - = 0.292L ~ 300mL
 - Therefore, give ~300cc of 3% saline to raise this patient's serum Na by 5mmol/L

Hyponatremia – by ADH

Classification	Causes	Urine Studies	Treatment
ADH-independent Mediated by too much water intake, compared with salt/solute	Beer potomania (Uosm can be <100 or 100-300) Psychogenic polydipsia D5W Infusions	Una<20 Uosm<100 (ADH off)	Fluid restrict Salt/urea tabs (increases solute load AND will increase water diuresis)
Appropriate ADH	True hypovolemia (burns, bleeds, GI losses, pancreatitis) Decreased effective circulating volume (cirrhosis, CHF, hypoalbuminemia = third spacing fluids)	Una<20 (Aldo on) Uosm>300 (ADH on)	IVF Resuscitation Diuretics “Treat underlying cause where possible”
SiADH (inappropriate ADH)	Nausea/vomiting/pain Drugs- TCAs, SSRIs, carbamazepine, ecstasy/MDMA, Thiazide Pneumonia/Lung cancer (small cell, classically)/CNS disorders Adrenal insufficiency, hypoT4	Una>40 (Aldo off) Uosm>300 (ADH on)	Fluid restrict Salt/urea tabs Furosemide SGLT2i Tolvaptan “Treat underlying cause where possible”

Hyponatremia – 3 Special Situations

Isotonic hyponatremia (sOsm 280-295)= Pseudohyponatremia

- hypertriglyceridemia
- paraproteinemia (multiple myeloma)
- obstructive jaundice

Hypertonic (sOsm >295) hyponatremia

- hyperglycemia
- mannitol
- Immunoglobulins (IVIG)

Hyperglycemia correction (approximate, values in mmol/L)
 $\text{Sodium (corrected)} = \text{Sodium (measured)} + \text{glucose} / 4$

Hyponatremia in the dialysis population: they're drinking too much water in between sessions! Treatment: fluid restriction (not salt restriction)

Over correction

- Avoid rapid correction in chronic HypoNa to prevent osmotic demyelination
 - Risk factors for demyelination: hypokalemia, malnutrition, alcoholism, liver disease, low starting Na
- Tip: “If urine output exceeds 150 ml/hr page MD”
 - Indicates that ADH has turned off and patient will suddenly start peeing free water (dilute urine) thus raising serum sodium faster than expected
- IF overcorrected:
 - DDAVP
 - D5W
 - Call nephro

Risk Factors	Change in Na per Day for chronic HypoNa	
	Target	Maximum
None	6	8
Any risk factor	4	6

Hyponatremia MCQ

A 64-year-old man with schizophrenia with history of psychogenic polydipsia presents with serum sodium of 112 mEq/L and new-onset confusion. He receives two 100 mL boluses of 3% NaCl. Over the next 6 hours, his sodium rises to 124 mEq/L. What is the most appropriate next step?

- A. Administer another 100 mL bolus of 3% NaCl
- B. Start free water restriction and monitor sodium closely
- C. Administer desmopressin and D5W to relower sodium
- D. No intervention needed if symptoms improve
- E. Start furosemide to promote free water clearance

Hyponatremia MCQ

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His sodium rose **12 mEq/L in 6 hours**, which exceeds the safe correction threshold (≤ 8 -10 mEq/L in 24h assuming this was acute). Desmopressin (to halt water diuresis) + D5W (to lower sodium) is used to prevent **osmotic demyelination syndrome (ODS)**.

Management of Hypernatremia

Free Water Deficit

1. Calculate **Water Deficit**= % change in [Na] x TBW

- Where % change in [Na] = $\frac{\text{Serum Na} - 140}{140}$
- **Water Deficit**= % change in [Na] x TBW

TBW= Total Body Water

Female= 50% of Wt (Kg) or Wt (Kg) x 0.5

Male= 60% of Wt (Kg) or Wt (Kg) x 0.6

2. Determine rate of correction of water deficit:

- Acute (e.g. postop central DI, nephrogenic DI): Correct quickly (rare)
- Chronic (e.g. dementia LTC patients): Correct water deficit slowly: maximum 0.5 mmol/L per hour/ 12 points over 24 hours (Avoid cerebral edema!)
 - **IF POSSIBLE USE ENTERAL ROUTE (If NG, give free water flushes)**
 - Large volumes of D5 → hyperglycemia → glucosuria & a solute diuresis, worsening polyuria and hypernatremia

Diabetes Insipidus (DI)

- Clinical clues: polyuria + hypernatremia
 - (and polydipsia if patient is able to drink)
- Types of DI:
 - Central DI (brain can't make/secrete ADH)
 - Nephrogenic DI (kidney can't respond to ADH)
 - causes: Lithium, hypercalcemia, hereditary, resolution of obstructive nephropathy
- Diagnosis:
 - **Step 1: One of:**
 - **Labs: high serum Na with inappropriately low urine osmolality confirms diagnosis**
 - Water deprivation test: Urine osmolality does not rise appropriately despite rising serum osmolality/serum Na
 - Step 2: DDAVP (2-4 mg IV/SC) test to differentiate between central and nephrogenic DI

Diabetes Insipidus

	Diagnostic Criteria	Response to DDAVP
Central DI (e.g. postop neurosurgery, trauma)	- Serum Osm > 295, Na > 145 But Urine Osm is low, (< Serum Osm) After water deprivation test, urine osm remains low	<u>Responds!</u> Urine osm increases (by 50%) Urine output decreases
Nephro DI (e.g. Lithium toxicity, sickle cell nephropathy)	- Serum Osm > 295, Na > 145 But Urine Osm is low, (< Serum Osm) After water deprivation test, urine osm remains low	Complete nephro DI: no change Partial: outside the scope of this exam!
Psychogenic Polydipsia	Sosm<295, Na < 145, U osm <100 After water deprivation test, urine osm rises / concentrates appropriately	No change (<10% change)

****Psychogenic polydipsia patients do *not* develop hypernatremia**** (polyuria + hypOnatremia + polydipsia)

3. Hypo/Hyperkalemia

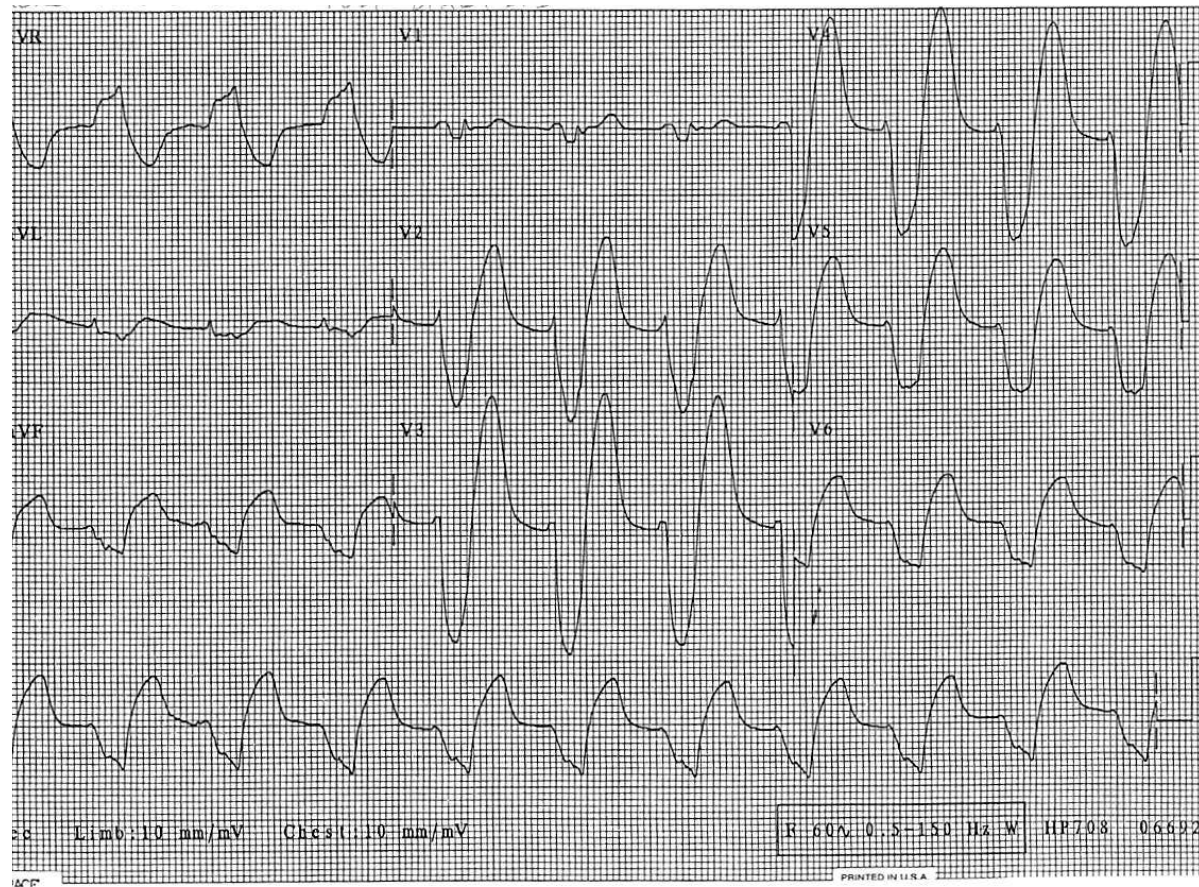


Potassium Question

A 24 year old female presents to your office with the following labs. She has a BMI of 17.0, and appears thin. Labs: Na 133, K 2.4, Cl 95, Hco3 30. Urine studies show: high urine K, low urine calcium, low urine chloride. Her vitals are 91/54 with a HR of 106 bpm. What is her diagnosis?

- a) Gittleman's syndrome
- b) Primary aldosteronism
- c) Bulimia Nervosa
- d) Barter's syndrome

Heart stopping!



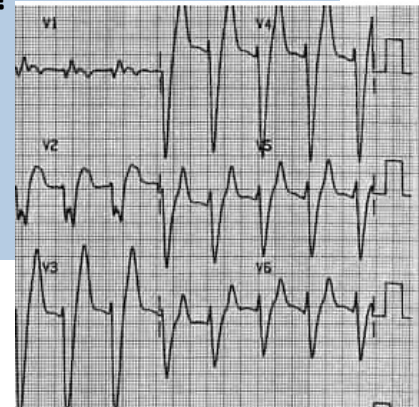
K=8.2!

Hyperkalemia Causes

- **Increased intake – unlikely to be sole cause**
 - PO/IV K supplement
 - Transfusions
- **Decreased excretion**
 - Decreased tubular flow: CKD, Volume depletion
 - Drugs- ACEi/ARB, NSAIDs, MRAs
 - Hypoaldosteronism: adrenal insufficiency, RTA type 4
- **Shifting**
 - Cell lysis- TLS, Rhabdo, burns
 - Metabolic- Acidosis, low insulin
 - Hyperosmolarity: glucose, mannitol
 - Familial hyperkalemic periodic paralysis
- **Factitious**
 - Fist clenching/ tourniquet
 - **Hypercellular blood** (ie- heme malignancies)
 - High WBCs, thrombocytosis
 - Do a VBG/ABG! – no tourniquet and no centrifuge
 - = no hemolysis in tube
 - Hemolyzed sample

Is it a medical emergency?

Peaked T waves
Flattened Ps
Prolonged PR
Brady-arrhythmias
Prolonged QRS



Hyperkalemia Treatment

Strategy	Specific Orders	Notes
Stabilize cardiac membrane	Calcium Gluconate IV (or CaCl)	Does NOT remove potassium
Stop Offending Meds	Stop RASi, MRAs, Calcineurin inhibitors, Septra	
Reduce Oral Intake	Dietary low K	
Shift K into cells	NaHCO ₃ (to correct acidosis) D50W + 10 units Insulin Regular	
Remove potassium (via gut)	K Binders - Lokelma (Rapidly exchanges K in gut for Na⁺ and H⁺) - Veltassa (Patiromer) (bind K in exchange for Ca) Kayexelate (polystyrene sulfonate - exchange Na ⁺ in stomach for H ⁺ , which is then exchanged in colon for K ⁺ and resin is excreted)	Do not use if bowel obstruction. Lokelma is \$\$\$.
Removes potassium (via kidney)	IVF (if hypovolemic) Furosemide (if overloaded)	Treat hyperkalemia with C-BIG K: C- calcium gluconate 1g IV – stabilize the heart B- beta agonist (Ventolin)/ Bicarbonate I-insulin (10 units insulin R IV) G- glucose (1 amp D50 BEFORE insulin) K- K removal: Furosemide/K binders/ dialysis
Remove potassium (via RRT)	Dialysis – need Hemodialysis	

Hypokalemia

First 3 Causes:

1. Decreased intake
2. Check Serum Mg
3. Shifting (usually causes acute hypokalemia)
 - Endocrine causes
 - ++Insulin, thyrotoxic periodic paralysis
 - Stress: ++ catecholamines
 - Acid/base
 - Metabolic alkalosis
 - Alcohol withdrawal
 - Hypothermia
 - Drugs
 - Amphetamines, antipsychotics (rare)

Emergency? respiratory muscle weakness, arrhythmias

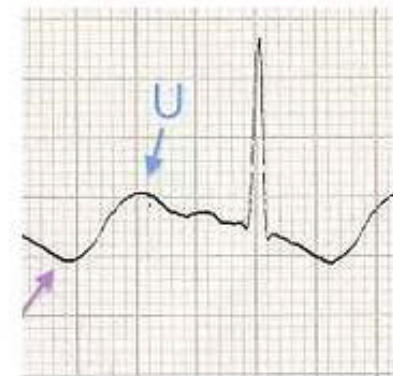
ECG changes

PR prolongation

TWI

ST depressions

U waves



Hypokalemia

Further Workup:

Metabolic Acidosis -> **Refer to RTA Slides** (Type I or Type II)

Metabolic Alkalosis -> **Refer to Metabolic Alkalosis Slides**

- HTN -> Check Aldo:Renin -> Liddle/licorice/steroids vs. Primary aldosteronism (PA) vs. Renal Artery Stenosis (RAS)/Reninoma
- Low BP -> GI Causes vs. Barter's/Gittleman's vs. Diuretics

Potassium Question - Answer

A 24 year old female presents to your office with the following labs. She has a BMI of 17.0, and appears thin. Labs: Na 133, K 2.4, Cl 95, Hco3 30. Urine studies show: high urine K, low urine calcium, low urine chloride. Her vitals are 91/54 with a HR of 106 bpm. What is her diagnosis?

- a) Gittleman's syndrome
- b) Primary aldosteronism
- c) **Bulimia Nervosa**
- d) Barter's syndrome

Nausea/vomiting/bulimia (or anorexia, binge-purge subtype)

Will present with

- low K
- alkalosis
- low normal BP
- Urine chloride LOW! (as opposed to Barters & Gittlemans, that have high urine chloride)

Overview

1. ~~Acid Base~~
2. ~~Hypo/Hybernatriemia~~
3. ~~Hypo/Hyperkalemia~~
4. Hypertension
5. Glomerular Disease
6. AKI
7. CKD Complications
8. DM2 and CKD 2022

- BONUS Nephro Nuggets
 - Acronyms used
 - “Lumps” in the kidney
 - ADPKD
 - Choosing Wisely
 - Nephrolithiasis
 - HTN (practical)
 - When to biopsy
 - Urine lytes/Hypokalemia (bonus)
 - Hematologic malignancies and the kidney
 - Landmark trials
- Bonus MCQ Cases

4. Hypertension



HIGH Yield for studying



Pearl- Exam Targeted Tips/ Trap Avoidance



Hypertension Question - 2026

A 55 year old male with no past medical history, non smoker presents for routine check up. BPTru is 135/85, hr 76, T 37, 99% oxygen. He does ambulatory home monitoring with average BP 132/85. Routine labs show eGFR 72, ACR 1.8, A1C 5.6, Lipid profile Framingham Risk Score 15%. His father and brother had hypertension well controlled with losartan. At what BP threshold would you start him on anti hypertensive pharmacotherapy?

- a. Systolic > 130
- b. > 130/80
- c. > 140/90
- d. >160/100

Blood Pressure Measurement



Office BP

Not Ideal for Diagnosis

Not Ideal for Monitoring

High out-of-office BP

Normal out-of-office BP

High Office BP

Sustained HTN

White Coat HTN

Normal Office BP

Masked HTN

Normal BP



Home BP

Acceptable for Diagnosis

Standard for Monitoring



Ambulatory BP

Gold Standard for Diagnosis

- *Validated Cuff (Heart Level)*
- *Reduce Stress (people, lights, substances, exercise)*
- *3 readings, 1 min intervals*

Optimal blood pressure measuring technique:

- ✓ Sitting position
- ✓ Back supported
- ✓ Arm bare and supported
- ✓ Use a cuff size appropriate for arm
- ✓ Middle of the cuff at heart level
- ✓ Lower edge of the cuff 1 cm above elbow crease
- ✓ See patient rest in calm room during rest period and between measurements
- ✓ Legs uncrossed
- ✓ Feet flat on the floor



Proper preparation:

- Quiet and calm room
- No caffeine, tobacco, or exercise in the 30 minutes before the measurement

Appropriate equipment:

- Validated sphygmomanometer device (see <https://hypertension.ca/public/>)
- Recommended devices (not all a pharmacist)
- Check range of cuff sizes to select the appropriate size



Standardized protocol:

- If feasible, observe a rest period of 5-10 minutes before measurements
- Measure BP 3 times at 1-minute intervals
- Average and record all values

* This slide was created by Dr. Sachin Pasricha, but IMR has permission to use/adapt.

Hypertension: Diagnosis

2025 Primary Care Guidelines

Note:

Does note apply to pregnancy or hyperacute/inpatient acute stroke

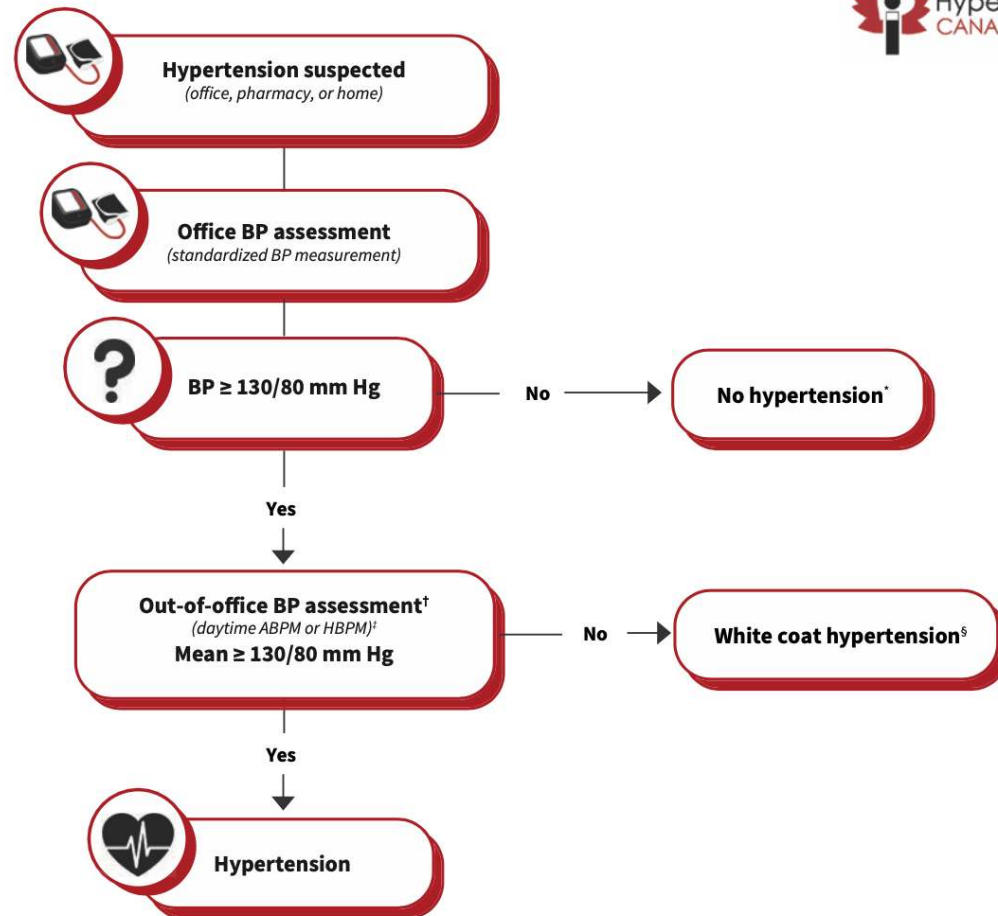
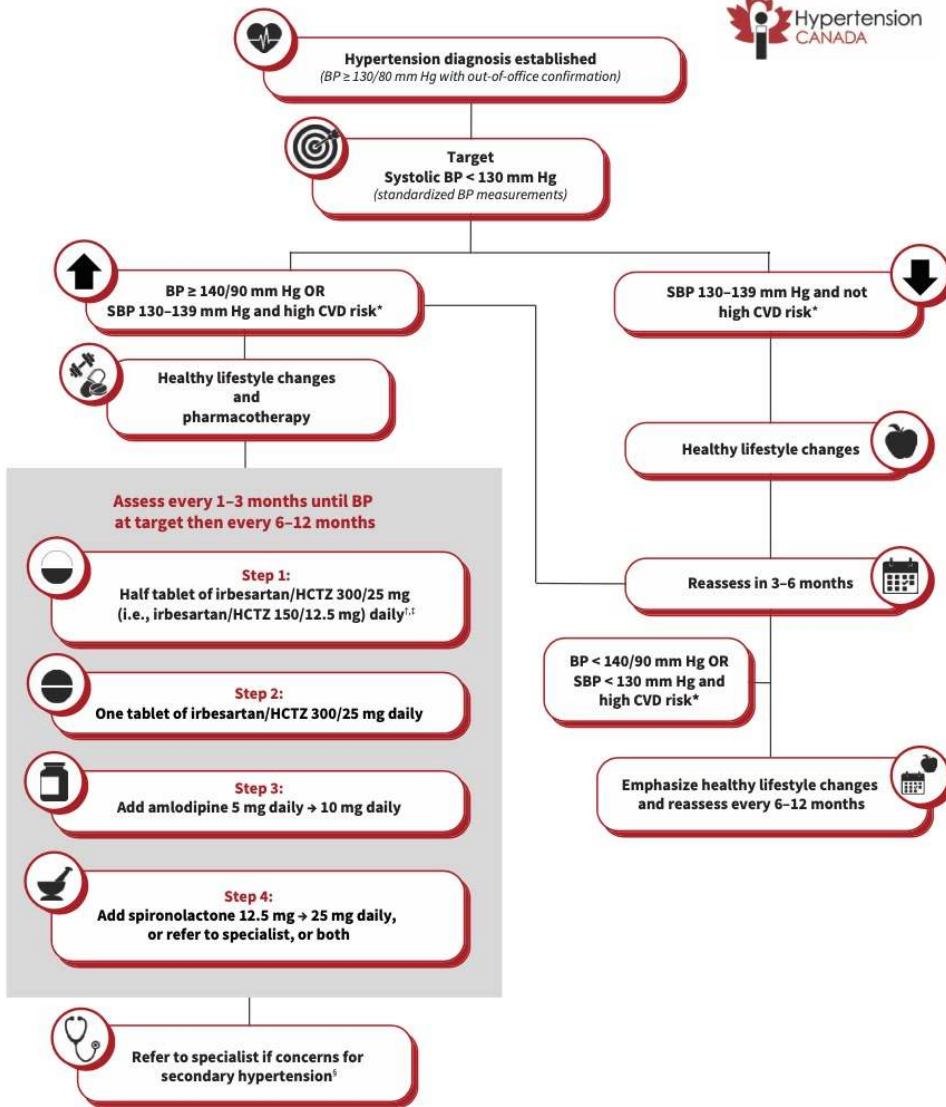


Figure 2: Suggested algorithm for the diagnosis of hypertension in primary care. Note: ABPM = ambulatory blood pressure monitor, BP = blood pressure, HBPM = home blood pressure monitor. *Continue yearly or opportunistic BP monitoring. Perform out-of-office BP assessment if masked hypertension is suspected. †If unavailable or infeasible, repeat office BP assessment at a separate visit. ‡Use Hypertension Canada–recommended or other validated home BP devices. §Continue yearly or opportunistic BP monitoring.

Hypertension – Routine Workup

(HTN 2020 Guidelines, no change in 2025)

- Routine Labs
 - Lytes (Na, K), Creatinine, Fasting Glucose and/or HbA1C, lipid profile (fasting or **non-fasting**), urinalysis
 - Urine ACR
 - B-HCG if potential for pregnancy
- ECG- 12 Lead
 - Echocardiogram **not routinely recommended** – perform if suspect LVH, systolic/diastolic dysfunction, CAD
 - HTN + evidence of heart failure – should assess LVEF (echo or nuclear test)
- All patients being treated for HTN should be monitored for new appearance of DM (as per CDA guidelines)
 - Also monitor lytes, creatinine and lipids – frequency determined by clinical picture



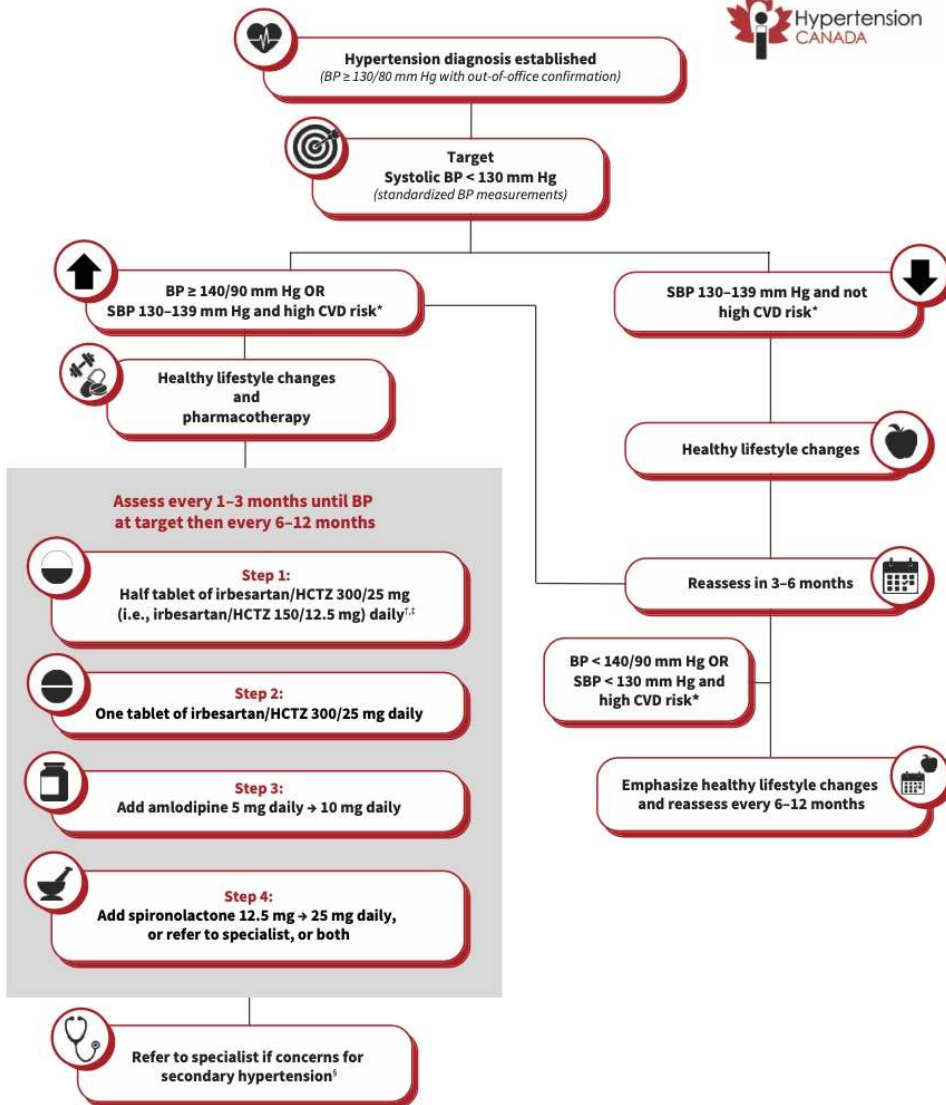
Hypertension 2025

Who to Treat

BP ≥ 140/90 OR
SBP ≥ 130 + High CVD Risk

High CVD Risk :

- Cardiovascular disease (established CAD, PVD, cerebrovascular disease, heart failure)
- Diabetes (type 1 or 2)
- CKD (eGFR <60 ml/min/1.73m² or albuminuria ≥ 3mg/mmol)
- 10 year Framingham Risk Score ≥ 20%
- Age ≥ 75 yrs



Hypertension – Initial Treatment

Lifestyle Measures

- A (Aerobic Exercise) moderate 150-300 min/week
- B (BMI) of 18.5-24.9
- C (Consume) <2 drinks EtOH/week, smoking cessation
- D (Diet) – DASH diet, <2000 mg sodium/day, supplement K if safe / not hyperkalemic
- E (Environment) – stress reduction, relaxation, CBT

Pharmacotherapy Key Points

Start with a combination RASi/Thiazide-diuretic

3rd Agent: CCB

4th Agent: Spironolactone + **Secondary HTN** Workup
- Spironolactone = Best 4th Agent (PATHWAY-2 Trial)

Target is <130 systolic

(HTN Canada does not aim to target to <120 mmHg –
i.e. does not use results of SPRINT Trial)

Follow-up q1-3 months until controlled, then q6-12
months

We anticipate you will ask:

1. Why no diastolic target?

- “We do not recommend a specific diastolic BP target, given evidence that adults with a systolic BP < 130 mm Hg are at relatively low cardiovascular risk even when diastolic BP is 70–90 mm Hg.”

2. Why not go to SBP <120 (as in KDIGO, SPRINT for instance) for those with high CV risk?

- “Rather than recommending the more intensive BP target of systolic BP < 120 mm Hg, we chose to recommend a more conservative systolic BP target of < 130 mm Hg, given that research- quality BP measures are generally 5–10 mm Hg lower than BP measures in the routine clinical care setting; modern-day RCTs studying intensive BP control included only people with baseline systolic BP ≥ 130 mm Hg; most participants in large RCTs targeting a systolic BP < 120 mm Hg did not achieve this target; and feedback from primary care providers and patients supported the recommendation “
- This is a PRIMARY CARE GUIDELINE – you will get oral scenarios with a clear SPRINT candidate where tight BP control is recommended on exam so do not define treatment by rigid guideline (see below)

3. What about a population that has a Canadian subspecialty guideline with a different target?

A/ IMR approach: Stick to the Subspecialty guideline, see next slide.

[Hypertension 2025 is a PRIMARY CARE guideline, the 2020 Guidelines are still valid / live]

SPRINT Trial NEJM 2015

- RCT of intensive BP control (SBP<120 versus standard <140)
 - Intensive arm reduced all-cause mortality, and composite primary outcome
- INCLUDED:
 - >50y **AND** SBP 130-180 **AND** one or more of the following:
 - Clinical or subclinical cardiovascular disease
 - CKD (non diabetic, proteinuria < 1g/day, GFR 20-60 ml/min)
 - Estimated 10 year global cardiovascular risk ≥15%
 - Age ≥ 75
- Excluded
 - Diabetes, history of stroke, GFR < 20, proteinuria > 1g/day, GN, PKD
 - Contraindications: non-adherent, secondary htn, life limiting disease, standing SBP < 110

2026 Q) What if I get a question about a "SPRINT CANDIDATE" on MCQ??? What do I pick? SBP 120 [SPRINT] or 130 [HTN Canada Primary Care]

IMR Answer) Be very sure it's a sprint candidate and they aren't tricking you (eg proteinuria, past stroke exclusions, severe frailty). If really a SPRINT candidate go for 120.

Hypertension Goals: Special Populations

Special population	Blood pressure goal
Acute Post ischemic stroke with thrombolysis (see Neurology Lecture for more)	180/105 (*not 185/110 - Stroke Best practices Guideline states for 24h post tPA target BP <180/105 so in order to give tPA In ED target <180/105)
Acute Post ischemic stroke without thrombolysis	<220/120
Ischemic stroke – chronic management	<140/90 Stroke best practices, <130 lacunar strokes <130 Hypertension 2025
Hemorrhagic stroke – acute management	Goal 140-160
Hemorrhagic stroke – chronic management	130/80
History of spontaneous ICH – chronic management	130/80
CKD	<130 for most KDIGO: Consider <120 (non DM, proteinuria<1g, non HD, not transplant) IgA Nephropathy <120/70
PKD (polycystic kidney disease)	<110
Post renal transplant	<130
Pregnancy (see OB Med Lecture)	DBP <85 (start treatment if SBP >140 or DBP >90)
Diabetes	<130/80

Initial Treatment: Response to Therapy

- If response to therapy is not as good as expected, rule out:
 - Non-adherence
 - Action: simplify to daily dosing; use Single Pill Combination (SPC) if possible; multidisciplinary team approach
 - White coat HTN: use ABPM or HBPM
 - Inaccurate measurements
 - Interfering drugs or substances, e.g. NSAIDs, corticosteroids, cocaine
 - Diet: high salt intake
 - Secondary HTN
 - Encourage more patient responsibility/autonomy
 - Monitor BP, adjust dose, educate!!

Secondary Hypertension

- Drugs, drugs, drugs, drug withdrawal, and licorice
- Vascular – coarctation of aorta
- Renal (Liddle's, GN, CKD...)
- Obstructive sleep apnea
 - Screen: STOPBANG questionnaire
 - Workup/Mgmt: Sleep study/CPAP or BIPAP

Renovascular Hypertension

- Due to atherosclerosis
- Due to FMD

Endocrine Hypertension

- Pheochromocytoma (PCC)
- Hyperaldosteronism
- Thyroid disorders
- Hyperparathyroidism (hyperCa)
- Cushing's
- Acromegaly



HIGH Yield for studying



Pearl- Exam Targeted Tips/ Trap Avoidance

Renovascular, PCC, Hyperaldosteronism are highly testable / high yield.

Covered in Hypertension 2020 Guideline.

The workup and management are not revisited in the 2025 guideline which is a primary care guideline and makes suggestions to refer for secondary hypertension workup.



Renovascular Hypertension

Type	Who to Screen	How to Screen	Treatment
Atherosclerosis	Patients with hypertension+ 2 or more of: • Sudden onset or worsening HTN age >55 or <30 • Abdominal bruit • HTN resistant to ≥ 3 drugs • Increase in Cr $\geq 30\%$ with ACEi or ARB • Other atherosclerotic vascular disease, particularly in smokers or dyslipidemia • Recurrent pulm edema associated w/ Hypertension emergency	Any of: —Renal Doppler US, captopril renogram, MRA, CTA <i>Avoid captopril and CTA if renal GFR <60 ml/min/1.73m²</i>	Medical management first line – RASi with close followup esp if bilateral stenosis. Angioplasty and stenting could be considered if any of the following present • uncontrolled HTN resistant to <i>maximally tolerated pharmacotherapy</i> • progressive renal function decline • Acute pulmonary edema
FMD	Workup if Hypertension and 1 or more of: — Kidneys asymmetrical (>1.5cm difference) — Abdominal bruit <i>but no atherosclerosis risk factors</i> — Confirmed FMD in another vascular bed — Family hx of FMD	CTA or MRA If FMD confirmed: Screen vasculature from head to pelvis with either CTA or MRA (cervicocephalic lesions, intracranial aneurysms, lesions in other vascular beds)	Refer to a Hypertension Specialist Renal artery angioplasty without stenting - Stenting is not recommended due to risk of periprocedural dissection.

Secondary Hypertension: PCC

- Pheochromocytoma – Who to Screen
 - Paroxysmal, unexplained, labile, and/or severe ($\geq 180/110$) sustained HTN refractory to usual therapy
 - HTN + symptoms of catecholamine excess (headaches, palpitations, sweating, flushing)
 - HTN triggered by beta-blockers, MAO-Is, surgery, anesthesia, micturition
 - ***Incidental adrenal adenoma***
 - Hereditary causes– such as Von-Hippel-Lindau, MEN 2A or 2B, neurofibromatosis type 1
- How to Screen
 - Biochemical screening test first
 - YES ✓ – 24hr urine total metanephrines and catecholamines (and Creatinine to ensure adequate collection)
 - MAYBE – Plasma free metanephrines and free normetanephrines may also be considered
 - NOT ✗ urinary VMA
- Treatment
 - Surgery
 - Pre-operative alpha blockade with phenoxybenzamine
 - Avoid diuretics, and only use beta-blockers IF already alpha-blocked (i.e. tachycardia controlled)

Secondary Hypertension: Hyperaldosteronism

- Hyperaldosteronism – Who to Screen
 - 1) Spontaneous hypokalemia $K < 3.5$, or diuretic-induced severe hypokalemia, $K < 3.0$
 - 2) Hypertension resistant to ≥ 3 drugs
 - 3) An incidental adrenal adenoma **AND hypertension**
- How to Screen
 1. Serum Aldo:Renin Ratio* - see next slide
 2. Confirmatory testing – see next slide
 3. CT abdomen
 4. Adrenal vein sampling
- Treatment

Unilateral -> surgery

Bilateral or Unilateral w/ non-surgical patient -> MRA

Hyperaldosteronism:

Screening test + Confirmatory test

Screening test: Serum aldo:renin ratio

Cutoffs vary by lab assay and units used

Be aware of medications to hold prior to sending patient to lab.

False positives	Beta-blockers, alpha-2-agonists, oral contraceptives	Stop >4 weeks
False negatives	MRAs, K ⁺ sparing diuretics	Stop >4 weeks
	RASi, thiazide/loop diuretics, dihydropyridine CCBs	Stop >2 weeks

Position: Patient must be ambulatory 2 hrs (sitting, standing, walking) before test; and must be seated at lab 5-15min before lab draw.

Fix: Hypokalemia, salt load

Avoid: MRAs, ENaC-Inhibitors ex amiloride, RASi, diuretics, beta blockers

Avoid: OCP (False+ if using direct renin conc.), NSAIDs

Ok: Verapamil/Diltiazem, hydralazine, alpha blockers(ex prazosin)

Confirmatory testing – autonomous hypersecretion of aldosterone is confirmed if one of these 3 are met:

- 🔴 Plasma aldosterone to renin ratio > 1400 pmol/L/ng/mL/h (or > 270 pmol/L/ng/L), **with plasma aldosterone > 440 pmol/L**
- 🔴 Saline loading test (2L NS over 4h, measure plasma aldo afterward, abN if >280 pmol/L)
- 🔴 Captopril suppression test (give 25-50mg captopril to pt standing for 1 hr. Measure renin, aldo levels at 0, 1, 2 hrs post. If aldo is suppressed >30% post ingestion hyperaldosteronism unlikely.

Hyperaldosteronism

- In patients with **hyperaldosteronism**, definite adrenal mass, and eligible for surgery, must first perform **adrenal vein sampling** to assess lateralization of aldosterone hypersecretion prior to ordering adrenalectomy
- For patients with established primary aldosteronism and in whom surgery is an option: selective adrenal venous sampling **to differentiate unilateral from bilateral overproduction of aldosterone**

Adjunct BP Meds – Just Scan!

Category	Examples or suffix	Precautions
Beta blockers (BBs)	--olol	Erectile dysfunction, bradycardia, and caution in asthma/COPD
Loop diuretics	Furosemide, torasemide, bumetadine, ethacrynic acid	Hyponatremia, hypokalemia, gout, ototoxicity
MRAs	Spironolactone, eplerenone, finerenone-not common practice yet	Caution in CKD, gynecomastia, hyperkalemia
Epithelial sodium channel blockers	Amiloride, triamterene	Hyperkalemia, caution in CKD
Alpha-1 antagonist	-zosin	Orthostasis, priapism
Alpha-2 agonist	Clonidine	Headache, somnolence, withdrawal
Smooth muscle relaxant	Hydralazine	Headache, palpitations, tachycardia, drug-induced lupus
Nitric oxide-mediated vasodilators	Nitro patch, isosorbide dinitrate	Reflex tachycardia, headache, not in posterior MI or sildenafil (Viagra) use

Hypertensive Emergency



Elevated BP + evidence of end organ damage

- Pulmonary edema, cardiac ischemia, neurologic deficits, acute renal failure, aortic dissection, and eclampsia

Treatment:

- 1) Admit to Step down/ICU level of care for continuous BP monitoring
- 2) Lower BP by 20-25% **only** within first 1-2 hours
 - Lower the blood pressure by 25% in 1-2 hours and then to 160/100 over the next 2-6 hours, and then gradually to normal over few days
 - Certain conditions (ie. Aortic dissection, eclampsia) may require rapid control of BP whereas other conditions (ie stroke) require even slower lowering of BP (<15% lowering)
- 3) Use of IV medications to lower BP (rapid acting and easily titratable)
 - IV med options: IV labetalol, esmolol, nicardipine, hydralazine, nitroglycerin
- 4) Transition to PO medications and transfer to wards
- 5) Consider work up of secondary hypertension

Special Situations	Treatment "back pocket drugs"
Cocaine	IV Phentolamine + Benzodiazepines Practically if phentolamine hard to source at 1am on Friday= labetalol both α - & β -blockade
MI or CHF	IV Nitroglycerin No IV yet? Give 3 sprays Nitro to get started
Dissection	IV beta blocker (lower HR first, then BP), then nitroglycerin IV
Preeclampsia	IV labetalol or hydralazine CHEW IR Nifedipine 5mg capsule **
Scleroderma Renal Crisis	IV Enalaprilat or captopril

Hypertension Question- 2026

A 55 year old male with no past medical history, non smoker presents for routine check up. BPTru is 135/85, hr 76, T 37, 99% oxygen. He does ambulatory home monitoring with average BP 132/85. Routine labs show eGFR 72, ACR 1.8, A1C 5.6, Lipid profile Framingham Risk Score 15%. His father and brother had hypertension well controlled with losartan. At what BP threshold would you start him on anti hypertensive pharmacotherapy?

- a. Systolic > 130
- b. > 130/80
- c. > **140/90**
- d. >160/100

This question pertains to Hypertension 2025 PRIMARY CARE guidelines. This is not a patient with high CV risk (FRS <20, no diabetes, eGFR >60 etc.)

Although mean BP $\geq$ 130/80 defines diagnosis of hypertension now in Canada, the new threshold to initiate in a patient without high CV risk is at bp $\geq$ 140/90

** This is a departure from previous guidelines where threshold to treat in low risk patient was SBP >160, DBP >100!

Overview

1. ~~Acid Base~~
2. ~~Hypo/Hybernatriemia~~
3. ~~Hypo/Hyperkalemia~~
4. ~~Hypertension~~
5. Glomerular Disease
6. AKI
7. CKD Complications
8. DM2 and CKD 2022

- BONUS Nephro Nuggets
 - Acronyms used
 - “Lumps” in the kidney
 - ADPKD
 - Choosing Wisely
 - Nephrolithiasis
 - HTN (practical)
 - When to biopsy
 - Urine lytes/Hypokalemia (bonus)
 - Hematologic malignancies and the kidney
 - Landmark trials
- Bonus MCQ Cases

5. Glomerular Disease

Nephritic Syndrome (“proliferative”)

Nephrotic Syndrome (“non-proliferative”)

2021 KDIGO GN Guidelines

2022 KDIGO Hep C guidelines

2024 KDIGO Lupus Guidelines

2024 KDIGO ANCA guidelines

2025 KDIGO IgA Guidelines



GN Question - 2024

A 35 year female presented with bilateral upper and lower extremity arthritis and a malar rash. Her lab values show an elevated dsDNA and positive ANA. She is referred to you for an up-trending creatinine (Cr 60>90>150), and elevated proteinuria (PCR 2g/day). You opt to biopsy her and it shows lupus nephritis class 3 / 4. What are your treatment recommendations?

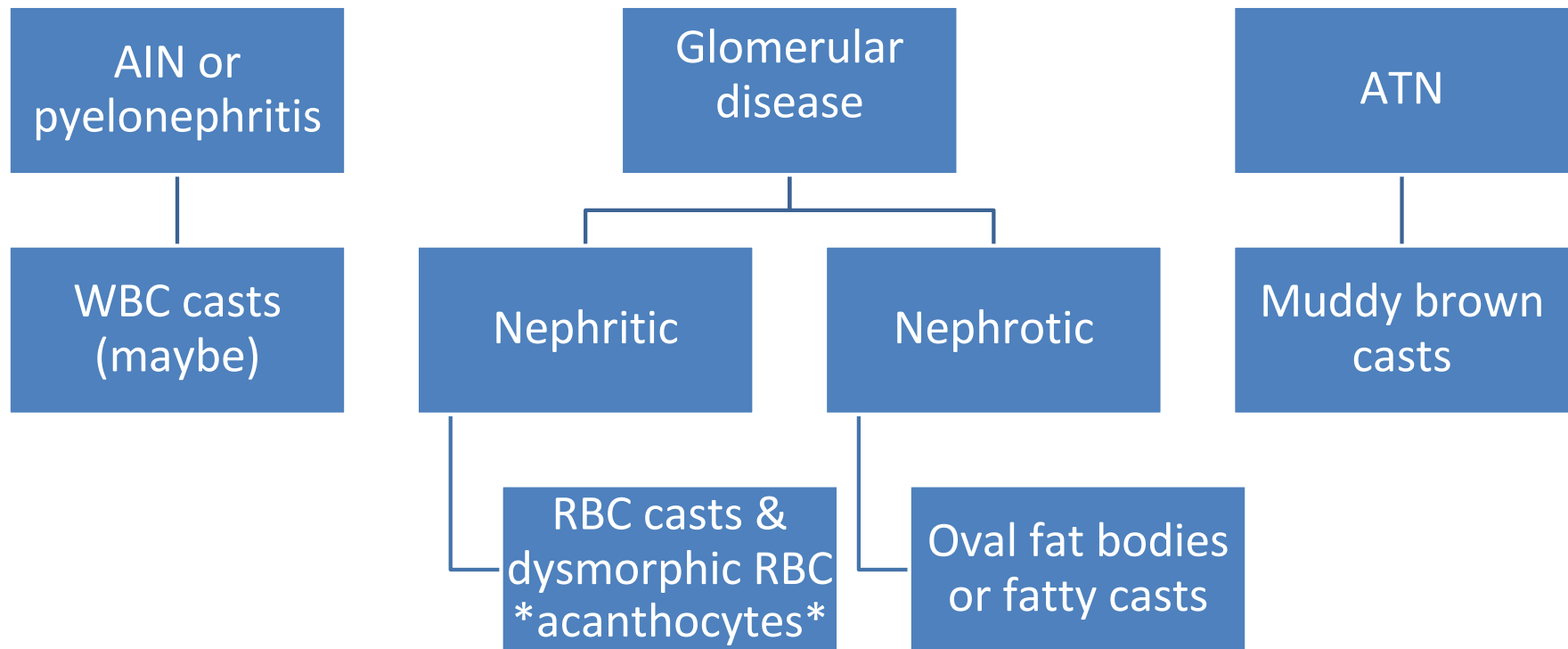
- a) Supportive therapy
- b) Tacrolimus for 6 months and monitor response to therapy
- c) IV solumedrol pulse + MMF therapy followed by prednisone taper + continuation of MMF
- d) IV solumedrol pulse only followed by prednisone taper

General approach to GN

- Patient will present with
 - Proteinuria (>500 mg/day)
 - Hematuria
 - AKI
- Classify
 - Nephritic: hematuria, proteinuria <3.5g/day, hypertension + symptoms of underlying cause
 - Nephrotic: proteinuria >3.5 g/day, low albumin, edema, high LDL
- Screen for symptoms:
 - Anti-GBM: hemoptysis
 - Vasculitis: constitutional, epistaxis, hemoptysis
 - Lupus: joint pain, rash
 - Malignancy: weight loss, B-symptoms
 - Drugs: thorough history of prescribed and non prescribed meds
- Work up: UA, 24 hour protein, anti-GBM, ANA, ds dna, RF, ENA, cryoglobulin, HIV/HepB/HepC, ANCA
 - More based on history: malignancy screening, CXR for sarcoid etc

Diagnosis: Biopsy is gold standard

Interpreting urine microscopy



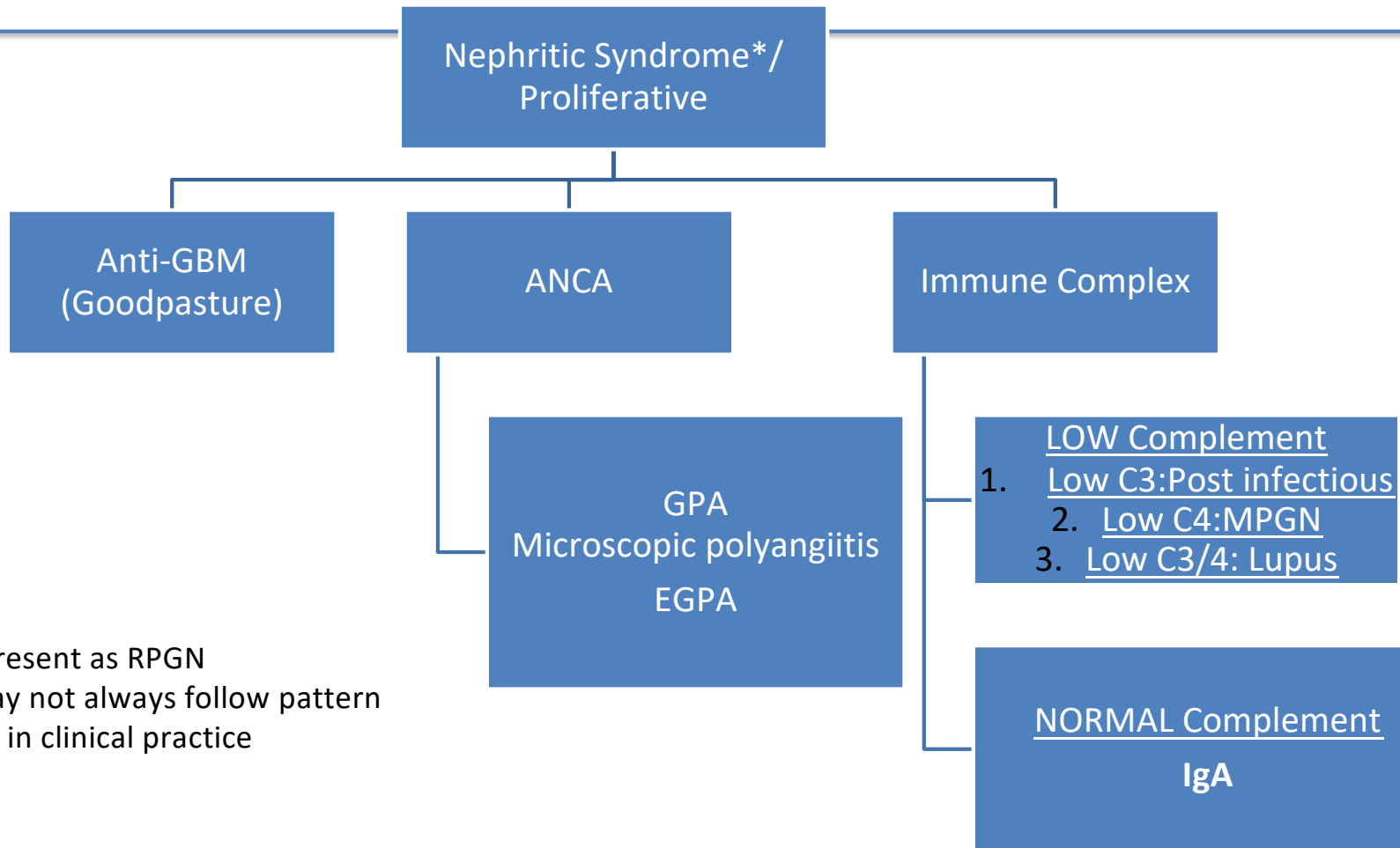
BONUS

Read on own

Kidney biopsy

- Indications: nephrotic/nephritic, especially when it will change management
- Risks:
 - Most common complication: bleeding (hematuria is expected), pain at biopsy site
 - Need for transfusion 1%, requirement for intervention to control bleeding < 1%, death 0.06%
- Medication management
 - Ideally would be best to stop antiplatelets 5 days before and resume 5 days post biopsy. But research shows that patients continuing asa at time of biopsy did not report increased risk of bleeding
 - Anticoagulation: involve a specialist! Individualize management
 - Warfarin: allow INR < 1.5, * bridge with heparin depends on patient
 - IV heparin: stop 6 hours prior and allow aPTT to normalize, resume 12-24 hours later
 - LMWH: stop the day prior and resume 48-72 hours later
 - DOAC: there is no data
 - DDAVP is sometimes given to prevent bleeding
 - Blood pressure should be controlled < 140-160 pre biopsy
- Contraindications:
 - UTI, uncooperative patient, bleeding diathesis, uncontrolled hypertension, poor visualization, infection
 - Relative: solitary kidney, hydronephrosis, small kidneys

Glomerulonephritis - Nephritic



*ANY can present as RPGN

**C3/C4 may not always follow pattern shown here in clinical practice

Anti-GBM disease

- Pathophysiology: auto-ab attacking basement membrane of glomeruli in linear fashion
 - **Anti-GBM** = renal limited disease.
 - **Goodpasture's syndrome** = renal + pulmonary disease
- Symptoms: AKI ± pulmonary hemorrhage
- Treatment: steroids + cyclophosphamide + PLEX
 - EXCEPTIONS: consider conservative treatment if 85-100% crescents, no pulm hemorrhage, treated w/dialysis at presentation [where risk of immunosuppression potentially outweigh benefits as pt has ESRD already]
- Steroids usually 6 months/ Cyclophosphamide 2-3 months
- No maintenance therapy!

ANCA Vasculitis *2024 new guidelines!

- Symptoms:
 - Constitutional symptoms
 - Arthralgias, rash
 - Sinusitis, asthma, pulmonary hemorrhage
 - AKI
 - Mononeuritis multiplex
- Serology: p-ANCA, c-ANCA (* can be ANCA negative on serology)
 - **don't forget to send off Anti-gbm since they can present similarly!
- Diagnosis: Biopsy
- Management:
 - If clinical presentation ✓ and serology ✓ → start treatment (immune suppressive therapy) without biopsy*

ANCA treatment *2024 guidelines!

- Induction: Pulse Steroids + Rituximab **OR** steroids + cyclophosphamide
 - Rituximab: preferred in children, fertility preservation, frail/old, relapse, PR3 ANCA
 - Cyclophosphamide: If life threatening and/or cr > 354 consider
 - Plex: cr >300, rapidly rising cr*, new need for HD, diffuse alveolar hemorrhage with hypoxemia, Co-positive with Anti-gbm
- Maintenance: while you taper steroids (PEXIVAS trial regimen)
 1. Continue rituximab, or 2. Azathioprine
 - Duration 18mos – 4yrs after induction of remission
 - Consider stopping after 3 months if they remain HD dependent AND no extra-renal manifestations
- Relapse: re-induction – preferably with rituximab
- Refractory: use the other induction agent, PLEX, increase steroids
- **Avacopan – a great alternative for patients with contraindications to steroids– used in both the induction and maintenance phase

Immune complex: Low C3

Low C3: Post-streptococcal/infectious GN

- Symptoms: 2–3 weeks post-infection (strep throat/strep cellulitis, chronic abscess, endocarditis, etc)
 - Presentation varies from
 - Microscopic hematuria to proliferative GN (red/brown urine, proteinuria, edema, Htn, AKI)
- Diagnosis: low C3, normal C4, +ASOT (70%), +anti-DNase B (90%)
 - Biopsy is rarely done
- Management: supportive care
 - Diuresis if edema, ACE/ARB if persistent proteinuria, treat infection if still active (e.g. cultured from any site)
 - Resolves in 3 – 4 weeks
 - Biopsy only if considering other glomerular disorders, or course varies from typical trajectory
 - i.e. severe AKI requiring dialysis: this is uncommon with PSGN

Immune complex: Low C4

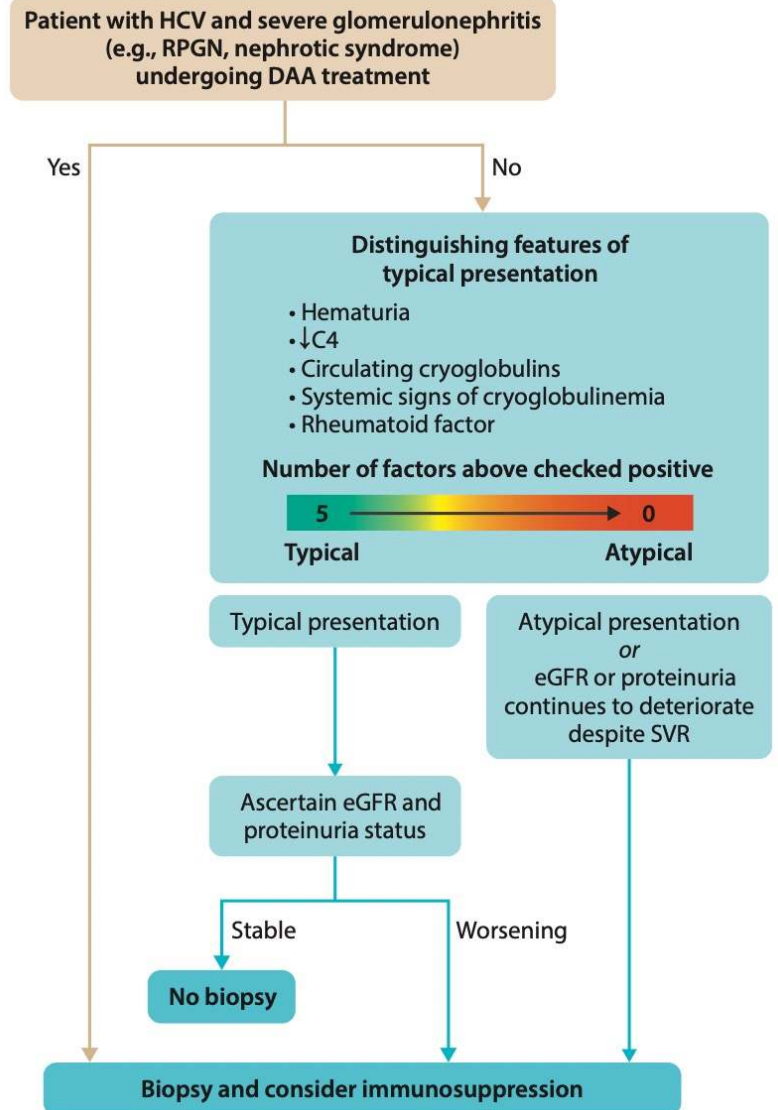
Low C4: MPGN (new pathologic classification based upon immunofluorescence, beyond scope of GIM)

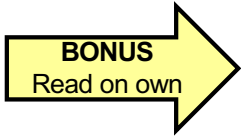
- Associated with: **Hep C, HIV, cryoglobulin, infections**
- 1. Immune complex mediated: supportive + treat underlying cause
 - Infection: Hep C, Hep B, cryoglobulin, Bacterial infections, malaria, Schistosoma
 - Autoimmune: SLE, Sjogren, RA
 - Monoclonal gammopathy or Fibrillary GN
- 2. Complement mediated: C3 GN or Dense Deposit Disease (DDD)
 - Treatment with MMF + steroids or eculizumab
- 3. Not immune complex or complement mediated: TMA, APLA, sickle cell
- Dx: on biopsy (IF is important!)
- See next slide for new HCV GN guidelines (KDIGO Nov 2022)

BONUS
Read on own

Hepatitis C and CKD – new KDIGO guideline 2022

- All patients with CKD should be screened for Hep C **at diagnosis of CKD**, and at time of **initiation of renal replacement therapy**
- All patients should be evaluated for HCV Therapy (if transplant candidate, coordinate timing w transplant centre)
 - Both pangenotypic regimens in your GI lecture can be even in CKD 5
- HCV + GN workup →
 - Note management nuances outside of scope of GIM... but FYI if RPGN or cryoglobulinemic flare KDIGO suggests Antiviral AND immunosuppression (Ritux) +/- PLEX
 - Or if patient has progressive GN or cryoglobulinemic kidney disease not responding to antivirals
“I would consult my colleagues in nephrology / hepatology / infectious diseases / heme...”





Immune complex: Low C3/C4

Lupus *New KDIGO 2024 guidelines

- All patients with SLE should have testing for GN at diagnosis, if suspicion of disease flare, and at regular intervals
 - Workup: Cr and eGFR, U/A, UPCR, anti-dsDNA and C3/C4
 - 24h urine protein if
 - U/A + acanthocytes, RBC casts or WBC Casts or U/A 2+ proteinuria or 1+ with low SG
 - UPCR >50 mg/mmol
 - Consider Kidney Biopsy if: Declining eGFR on serial testing **or** 24h urine protein ≥ 500 mg/d
 - General Management of patients with SLE and SLE GN
 - CV risk reduction
 - Pregnancy planning, contraception, ASA in preg
 - RAAS blockade if HTN or proteinuria ACEi
 - Infection prevention (influenza, pneumococcal vaccines, assess HBV/HCV/HIV status, Tb risk)
 - Sun protection
 - Adequate Ca/Vit D, Bone health risk assessment
 - Pregnancy** delay until ≥ 6 mos – 1 year after LN inactive
 - Safe meds: hydroxychloroquine, azathioprine, tacrolimus, cyclosporine, steroids.
 - Remember to add ASA before 16 weeks of gestation
 - Contraception – if active disease, APLA or hx thrombosis prefer progestin only options.
- Refer to Gyne to discuss fertility preservation if starting Cyclophosphamide.

Immune complex: Low C3/C4

Lupus *New KDIGO 2024 guidelines, New **ACR 2024 Guidelines**

TREATMENT: 6 classes of lupus nephritis.

Hydroxychloroquine for all, Blood Pressure Control for all

- Class **1 + 2**: supportive treatment (*unless nephrotic syndrome -> outside of GIM scope)
- Class **3 + 4**: Induction: steroids + one of: MPAA (MMF/MPA) OR cyclophosphamide
 - Other options: **steroids + [belimumab + MPAA] or [CNI + MPAA]**. Caution with CNI if significant renal impairment.
 - MPAA preferred in patients at high risk of infertility
- Maintenance: MPAA (other option: azathioprine or CNI) + taper steroids

Consider Induction with triple therapy (American College of Rheumatology)

Especially if nephrotic range proteinuria

MPAA = Mycophenolate Acid Analogues
GC = Glucocorticoids

Immune complex: Low C3/C4

Lupus *New KDIGO 2024 guidelines, New ACR 2024 Guidelines

- Class 5:

- Non nephrotic range proteinuria: RAAS, immunosuppression only if warranted by extrarenal SLE manifestations
- Nephrotic range proteinuria: RAAS + steroids + one of (MPAA, cyclophosphamide, CNI, ritux, azathioprine)
 - ACR recommends TRIPLE THERAPY for new onset Class V with >1g/d proteinuria (GC+MPAA+CNI)
- If worsening AKI remember to rule out renal vein thrombosis!
- Monitor level of proteinuria, and assess for edema, dyslipidemia, thrombosis risk

- Lupus with TMA:

PLASMIC score= if moderate/high risk ≥ 5 points – start PLEX + GC pending APLA, ADAMSTS activity

- Low ADAMSTS activity ($\leq 10\%$) : “SLE associated TTP” – PLEX + GC + Rituximab
- Normal ADAMSTS activity, No APLAS: *evaluate for other TMA etiologies* – if 1° or 2° complement-mediated TMA consider eculizumab (see Heme lecture – e.g. atypical HUS)
- Normal ADAMSTS, +ve APLAS : “APLA nephropathy” – anticoagulate +/- plasma exchange

- Class 6:

- Advanced sclerosis: Conservative therapy

MPAA = Mycophenolate Acid Analogues
GC = Glucocorticoids

Immune Complex: Normal C3/C4

IgA Nephropathy – NEW KDIGO 2025 Guideline

IgA Nephropathy

- *Most GN worldwide (esp Asians > Caucasians, rare if African descent)
- Associated with: celiac disease, autoimmune disease, hepatitis, HIV, cirrhosis
- **Symptoms:**
 - **Various presentations:** microscopic hematuria, gross hematuria, proteinuria
 - **RPGN or nephrotic syndrome** – i.e. Flares with ANY infection (“syn-pharyngitic”)
- What is HSP? **Systemic** IgA vasculitis with arthritis, purpura, GI symptoms.
- Diagnosis: *can only be diagnosed with biopsy*
 - Kidney biopsy should be considered in all adults with proteinuria >0.5g/d if IgAN is a possible diagnosis and biopsy is not contraindicated.
- Prognosis: use IgA Prediction Tool (calculator)
 - Low risk of progression: <0.5/day, BP controlled, low MEST-C Score
 - High risk: Over proteinuria and/or high Cr (15-25% 10-year ESRD-risk)

IgA Nephropathy (Cont'd)

Highlights from 2025 KDIGO

- Routine Treatment:
 - BP management $<120/70$,
 - **ACEi/ARB if proteinuria $> 0.5\text{g/day}$** (w/ or w/o HTN) OR DEARA (dual endothelin AngII receptor antagonist - e.g. sparsentan – not available in Canada)
 - **SGLT2i if proteinuria $> 0.5\text{g/day}$,**
 - Target proteinuria to $<0.5\text{ g/day}$,
 - Lifestyle modification (exercise, smoking cessation, weight loss)
- High-risk Patient treatment (typically $>0.5\text{ g/day}$ protein)
 - Immunosuppressive therapy
 - Budesonide PO (nefecon) – in guidelines, not available in Canada
 - **Oral low dose steroids (prednisone 0.5 mg/kg/d then taper over 9 mos)**
 - Other DMARDs studied in small trials, no consistent benefit
 - Tonsillectomy in a Japanese patient, Hydroxychloroquine if Chinese patient
 - Refer to nephrology to consider in a Clinical trial

Rapidly progressive IgAN, IgA Vasculitis, IgAN with nephrotic syndrome etc. well beyond GIM scope.

Nephrotic Syndrome:

- Nephrotic range proteinuria (>3.5g)
- Edema
- hypoalbuminemia
- Dyslipidemia & lipiduria

Nephrotic Syndrome



Membranous? Screen for malignancies based upon patient characteristics. Usual age-based screen PLUS by pretest probability (smoker? fam hx? Iron def anemia? Note KDIGO comments to consider PSA for men >50.)

Minimal change

1°

Secondary
Heme Cancer-
Hodgkins & Leukemias
Drugs- **NSAIDs, COX2i, Li,**
Infections (rare),
including TB

FSGS

1°

Secondary
Infxn: **HIV, parvoB19, EBV**
Drug: heroin, pamidronate, anabolic 'roids
Hyperfiltration: **obesity**, single kidney, reflux nephropathy

Membranous

1°

NEW
PLA2R and THSD7A



Secondary
SLE (Class V)
CANCER- solid tumors> heme malignancy (CLL)
Infection: **HBV, HCV, Syphilis, HIV**
Drugs: **NSAIDs, anti TNFs, gold, penicillamine**
Sarcoidosis

Other (nodular)

Amyloid/ MGRS*
Diabetes

Primary:
Membranous most common overall, FSGS most common in african Americans

* MGRS = monoclonal gammopathy of renal significance

Management of Nephrotic Syndrome

Diagnosis → usually* requires biopsy

*unless PLA2R antibody + normal renal function– can forego biopsy, but still do screening for 2° causes

General Care

- **Edema:** Na restriction + loop diuretics (1st line, add other diuretics prn)
- **Hypercholesterolemia:** Statins, heart healthy diet
- **Proteinuria:** RAASi, BP control
- **Thrombosis:** CONSIDER “Prophylactic” full-dose anticoagulation w warfarin if
 - **albumin <20-25 AND any of:** BMI>35 | Inherited thrombophilia (ex FVL) | NYHA 3-4 | Prolonged immobilization | Proteinuria >10g/day | Recent orthopedic or abdominal surgery
 - If high bleeding risk, KDIGO suggests ASA 81mg daily instead
 - DOACs = not studied in NS, highly –protein bound in blood so not ideal in this situation w hypoalbuminemia!

Definitive management

- Immunosuppression in most 1° Cases
 - Minimal Change, FSGS : Steroids
 - Membranous: Risk evaluation for progression to renal failure– low risk (wait and see)→ Higher risk - immunosuppressants
- Treat underlying cause if 2° Cause

GIM PEARLS from KDIGO 2021 for Nephritic/Nephrotic Disorders:

SMOKING CESSATION FOR ALL

DIET: NA <2g, heart healthy, **adequate protein if Nephrotic range proteinuria (0.8-1g/kg/d)**



HTN: Target <120 if possible / safe
(e.g no orthostasis, not frail... they comment “In practicality 120-130 in most pts w GN disease”)

VACCINATE: pneumococcal, influenza, SHINGRIX, COVID-19 mRNA vaccines, Hep B if not immune, RSV if age >60 (RSV approved for this indication but not publicly funded yet in Ontario)

PRE-IMMUNOSUPPRESSION:

TBST, Hep serologies, HIV
Strongyloides if endemic risk.

ON TREATMENT:

Contraception, Gonadal protection
Bone health [see endo lecture]

TMP-SMX on high dose steroids, Ritux or Cyclophosph

Buzzword Infections and GN

- HCV renal disease: MPGN +/- cryoglobulinemia
 - Classically low C4 (but not always); high RF
 - Clinically: nephritic/proliferative picture, palpable purpura, arthralgias, weakness, peripheral neuropathies
- HBV renal disease: membranous, MPGN, **polyarteritis nodosa (PAN)**
 - 1-5% of patients with HBV develop PAN (think of this if stem points to PAN)
 - Clinically: pre-renal AKI, **renal infarcts with PAN**
- HIV-associated nephropathy (HIVAN): collapsing FSGS
 - Clinically: Black males, advanced HIV (CD4 <200), nephrotic, high Cr
 - Treatment: ART and RAS blockade

GN Question

A 35 year female presented with bilateral upper and lower extremity arthritis and a malar rash. Her lab values show an elevated dsDNA and positive ANA. She is referred to you for an uptrending creatinine (cr 60>90>150), and elevated proteinuria (PCR 2g/day). You opt to biopsy her and it shows lupus nephritis class 3 / 4. What are your treatment recommendations in addition to hydroxychloroquine?

- a. Supportive therapy
- b. Tacrolimus for 6 months and monitor response to therapy
- c. IV solumedrol pulse + MMF therapy followed by prednisone taper + continuation of MMF**
- d. IV solumedrol pulse only followed by prednisone taper

Treatment of lupus nephritis class 3 / 4

1. Steroid pulse followed by prednisone taper
2. MMF or cyclophosphamide
3. Hydroxychloroquine + BP control for all

Note: If there is an option that also includes Belilimumab, then choose this option (i.e., TRIPLE THERAPY).

6. Acute Kidney Injury



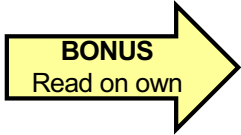
Common causes of AKI

- Pre-renal
 - Hypovolemia (dehydration, hemorrhage, vomiting, rhabdomyolysis, tumor lysis syndrome)
 - Cardiorenal/heart failure
 - Sepsis, anaphylaxis
 - NSAIDS, contrast, RAASi
- Intra renal
 - AIN
 - Nephrotoxic drugs
 - GN, autoimmune disease, vasculitis
 - Thrombosis
 - Sick cell
 - Abdominal compartment syndrome
 - Rhabdomyolysis (pigment nephropathy)
- Post renal
 - Obstructive uropathy

Acute interstitial nephritis [AIN]

*Great oral scenario!

- Causes
 - Drugs (#1 cause), can see as manifestation of DRESS syndrome (Drug Reaction w Eosinophilia & Systemic Symptoms)
 - Infections
 - Systemic illnesses: “The S’s”: SLE, sarcoid, Sjogren’s, IgG4 disease
- Presentation
 - Timing: 1 week to weeks/ months following. **If previous exposure to drug, can occur sooner**
 - Acute renal failure, sometimes with fever and/or rash
 - Hematuria, **non-nephrotic** range proteinuria
 - Pyuria (WBC in urine) and WBC casts (**not always**)
 - Eosinophils may be increased in blood and/or present in urine (**not always**)
- Common drugs (*not* a dose-dependent effect):
 - Anti-inflammatory: NSAIDs, COX-2 inhibitors
 - Anti-biotics: penicillins, cephalosporins, sulphonamides, rifampin, cipro, septr
 - Anti-gout: allopurinol
 - Anti-acid: PPI’s
 - Anti-edema: loop diuretics, thiazides
 - Immunotherapy
- Treatment
 - Remove offending agent.
 - If severe renal failure, often biopsy – especially if requiring dialysis!
 - In **select** cases, consider empiric glucocorticoids if severe renal failure, but also arrange for bx.



Contrast Induced Nephropathy

Contrast-induced nephropathy (CiN)

- PREVENTION: None
 - Hold metformin, RAS , diuretics, SGLT 2, avoid NSAIDS prior to the study
 - Ensure euvolemia (3 ml/hg/hr 1 hour before procedure, 1ml/kg/hr 6 hours after procedure for CKD)
- TIMING: 1 – 3 days post contrast load
- COURSE: nonoliguric, most self resolve in 1 – 2 weeks
- Treatment: supportive

Cholesterol emboli syndrome/ atheroembolic disease

- PREVENTION: none
- TIMING: subacute, step-wise decline – takes weeks
- COURSE: early on: urine eosinophils, elevated ESR, peripheral eosinophilia, decreased C3/C4
 - Urine R+M otherwise unremarkable
 - Cutaneous manifestations: livedo reticularis, **blue toes**

Rhabdomyolysis

- Symptoms: Muscle pain, weakness, cola coloured urine
 - Malaise, fever, tachycardia, nausea, vomiting in some
- Physical exam: muscle tenderness and swelling
- Lab findings: high CK, + RBC on dipstick, - blood on microscopy
 - AKI from hypovolemia and third spacing
 - Hyperkalemia and hyperphosphatemia from damaged muscle cell rupture
 - Hypocalcemia from deposition of calcium salts in damaged muscle
 - Uricemia from rupture of damaged muscle cells
 - Metabolic acidosis
- Treatment:
 - IV fluids
 - Management of electrolytes
 - Do not treat hypocalcemia

RHABDO causes – good for GIM to think about!

Trauma / crush injury / compartment syndrome / severe exertion

Toxin/Drugs (Statin, illicit drugs, NMS, malignant hyperthermia)

Metabolic/ mitochondrial disorders

Autoimmune
DM/PM

Immune Mediated Necrotizing myopathy / Anti HMG-COA reductase

Less likely – infections causing myositis (eg flu, legionella don't usually get to rhabdo levels)

Cardiorenal syndrome

“Any acute or chronic problem with heart or kidneys that may result in chronic problem in the other”

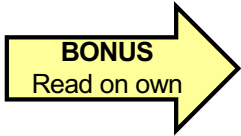
- 5 different types of Cardiorenal syndrome beyond GIM scope

Management

- “treat underlying cause”
- Improve cardiac function (LVAD and cardiac resynchronization therapy)
- Volume control (aggressive fluid removal - diuresis with furosemide)
- RAS blockade (*not associated with improvement of gfr)
- SGLT 2 inhibitors (avoid if eGFR <20 or fasting/shock/risk of DKA)
- Ultrafiltration (removal of isotonic fluid from venous compartment via semipermeable membrane)

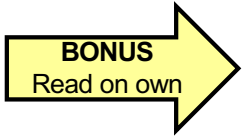
Hepatorenal syndrome (+see GI lecture)

- Clinical presentation:
 - Rise in serum creatinine with normal urine sediment
 - Tubular function is preserved (no/minimal hematuria or proteinuria!)
 - Urine sodium < 10
- Type 1 HRS:
 - Two fold increase in serum creatinine to a level greater than 221 $\mu\text{mol/L}$ over less than 2 weeks
- Type 2 HRS:
 - Gradual kidney decline with ascites refractory to diuretics
- Diagnosis: discontinue diuretics + antihypertensives, give IV albumin 1g/kg and you will not see an improvement
- Treatment: stop diuretics, restrict sodium and water
 - Treatment with albumin + vasoconstrictor (terlipressin is not available here, thus we give midodrine + octreotide)
 - Liver transplant or TIPS if candidate
 - Involve palliative care if not transplant candidate



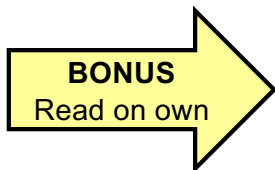
Causes of AKI in transplant recipient

- *****ACUTE REJECTION! (always suspect this)*****
 - Often varying degree of proteinuria, hematuria
 - Requires biopsy to diagnose
 - Immunosuppressive med levels can be **low**
- Medication induced (Cyclosporine, Tacrolimus)
 - Reversible, bland sediment, no blood or protein, immunosuppressive med levels will be **high**
- Infection (bacterial - anytime, BK virus – later complication)
 - WBC in urine, mild proteinuria/hematuria, tenderness over graft if pyelo
- Renal artery/vein thrombosis,
 - Usually early post-operatively. **Requires urgent ultrasound with dopplers of transplant kidney**
- Hydronephrosis
- Recurrence of primary disease
- PTLD (post transplant lymphoproliferative disorder – EBV associated)
- (Causes of AKI in native kidneys)



Causes of AKI in Cancer Patients

- Secondary to chemotherapy
 - Volume depletion from nausea, vomiting, diarrhea
 - Sepsis from immunosuppression
 - Chemotherapy related nephrotoxicity (cisplatin, carboplatin, mtx, gemcitabine, paclitaxel, doxorubicin, vincristine) → can be directly nephrotoxic or cause TMA, AIN, hypertension
- Secondary to malignancy spread
 - Tumor infiltration of the kidney (seen on renal imaging)
 - Obstructive nephropathy from retroperitoneal lymphadenopathy (seen on renal imaging)
- Hematologic emergencies
 - DIC (hemolysis)
 - TLS (high K, high uric acid, high phos!)
- Hypercalcemia
- GN (membranous nephropathy → will see nephrotic range proteinuria)
- Malignancy specific
 - Multiple myeloma → Cast chain nephropathy (will cause non albumin proteinuria)
 - Waldenstrom → hyperviscosity
 - Amyloid → fibril deposition (congo red positive + nephrotic range proteinuria)



Common Nephrotoxic Drugs

- **Contrast** (see AKI with Angiography slide)
- **ACEi / ARB**: pre-renal azotemia/ischemic ATN
 - Failure of efferent arteriole to constrict if volume contracted
 - Usually concomitant history of poor PO intake and/or GI losses
- **NSAIDs**: pre-renal azotemia/ischemic ATN, AIN, membranous nephropathy, minimal change disease
 - Prostaglandin inhibition → constriction of afferent arteriole (mechanism of pre renal)
- **Lithium**: tubular disease
 - Nephrogenic DI
 - Chronic tubular-interstitial disease (use amiloride)
- **Aminoglycosides/cisplatinum chemo**: toxic ATN/ tubular injury
 - Non-oliguric;
- **Vancomycin**: rare; mechanism unclear, associated with supratherapeutic levels
 - Continuous infusion of vanco associated with 53% reduction in odds of AKI (Flannery et al. June 2020)
- **PPI's**: Acute interstitial nephritis

Nephrolithiasis

	Calcium Oxalate (most common)	Calcium Phosphate	Uric Acid	Struvite	Cystine
Imaging	Radio-opaque	Radio-opaque	Radiolucent	Radio-opaque	Radio-opaque
Disease associations	-Common - Crohn's/ ileal disease: oxaluria	-type 1 RTA -hyperPTH	-heme disorders - ADPKD	-UTI -Proteus, Klebsiella	-congenital (Autosomal recessive)
Urine pH	variable	>6	<5	>7	<6.5
Treatment	- decrease Na and meat intake -oxalate restriction (limit Vitamin C) -thiazides (not if hypercalcemic) -do <i>not</i> limit Ca intake (will worsen oxaluria)	-decrease Na and meat intake -thiazides	-urine alkalinization (give K citrate) -allopurinol	-treat UTI	-urine alkalinization

All treatment involves ↑ oral fluids for urine output >2 L/ day
Tamsulosin for stones 0.6- 1 cm may help clear
Nifedipine do not appear to benefit stone clearance

Overview

1. ~~Acid Base~~
2. ~~Hypo/Hybernatriemia~~
3. ~~Hypo/Hyperkalemia~~
4. ~~Hypertension~~
5. ~~Glomerular Disease~~
6. ~~AKI~~
7. CKD Progression
8. CKD Complications

- BONUS Nephro Nuggets
 - Acronyms used
 - “Lumps” in the kidney
 - ADPKD
 - Choosing Wisely
 - Nephrolithiasis
 - HTN (practical)
 - When to biopsy
 - Urine lytes/Hypokalemia (bonus)
 - Hematologic malignancies and the kidney
 - Landmark trials
- Bonus MCQ Cases

7. Chronic Kidney Disease (CKD) Progression

2024 KDIGO CKD guidelines



...and this dish is totally potassium-free!

Classification of CKD

KDIGO CKD Definition-
abnormalities in either kidney structure or function **for >3 months**

- Albuminuria ACR >3 mg/mmol
- Urine sediment abnormalities
- persistent hematuria
- Electrolyte abnormalities due to tubular disorders
- Structural abnormalities detected by imaging
- Kidney Transplant hx.
- eGFR < 60mL/min/1.73m²

- In Adults with CKD – we recommend using a combination of creatinine and cystatin based gfr if available*
- Assess albuminuria at least annually
- Doubling of ACR or decline in eGFR >20% warrants evaluation

KDIGO: Prognosis of CKD by GFR and albuminuria categories

				Persistent albuminuria categories		
				Description and range		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				<30 mg/g <3 mg/mmol	30–300 mg/g 3–30 mg/mmol	>300 mg/g >30 mg/mmol
GFR categories (ml/min/1.73 m ²) Description and range	G1	Normal or high	≥90	Green	Yellow	Orange
	G2	Mildly decreased	60–89	Green	Yellow	Orange
	G3a	Mildly to moderately decreased	45–59	Yellow	Orange	Red
	G3b	Moderately to severely decreased	30–44	Orange	Red	Red
	G4	Severely decreased	15–29	Red	Red	Red
	G5	Kidney failure	<15	Red	Red	Red

Green: low risk (if no other markers of kidney disease, no CKD); Yellow: moderately increased risk; Orange: high risk; Red: very high risk. GFR, glomerular filtration rate.

Kidney Failure Risk Equation

kidneyfailurerisk.com

- Risk prediction tool to estimate absolute risk of kidney failure
 - Now provided on lab reports in Ontario if Cr, ACR ordered together
- 5 year KFRE of 3-5%
 - warrants a referral to nephrology
- 2 year KFRE >10%
 - warrants MCKC clinic
- 2 year KFRE > 40%
 - start prepping for dialysis



Delaying CKD progression: Other

- **Lifestyle:**
 - 150 min week/exercise, quit smoking optimize weight
 - Sodium <2 g/day; protein of 0.8/kg/day if CKD 3-5
- **Blood pressure** control
 - <130 for most
 - KDIGO: Consider <120 (non DM, proteinuria<1g, non HD, not transplant)
 - IgA Nephropathy <120/70
- Glycemic control if Diabetic
- **Dyslipidemia:** (**CKD**, not Dialysis patients)
 - >50 years + CKD 1-2: Statin.
 - CKD 3-5: statin or statin/ezetimibe.
 - For 18-49 in CKD-12: treat as general population (CAD, DM, stroke etc.)
- **Proteinuria:** <500 mg-1g/day. (See next slide).

Delaying CKD progression: CKD Meds

- **RASi** use in CKD 1-4 + albuminuria A2-3 **WITH/WITHOUT diabetes**
 - Can also use in patients with mild proteinuria with other indications (CHF, hypertension etc)
 - Measure BP, creatinine, potassium within 2-4 weeks of starting/increasing RASi
 - Hyperkalemia can be managed with binders (instead of stopping RASi)
 - Continue RASi unless creatinine rises >30% within 4 weeks
 - **Continue RASi even with GFR falls below 30**
- **SGLT2i** use in **DM + Non-DM CKD**
 - “recommend” if GFR ≥ 20 AND Any of (ACR ≥ 20 mg/mmol and/or CHF and/or DM)
 - “suggest” in anyone with gfr 20-45 *even if no diabetes or microalbuminuria*
 - “reasonable to continue” SGLT2i until dialysis or transplant even if GFR falls below 20 if tolerated
 - Prevent **composite** of decline in eGFR, progression to ESRD, kidney death, all cause mortality, nonfatal MI, hosp for HF
- **Nonsteroidal MRA (finerenone) used in CKD-DM + gfr >25** + ACR >30 despite maximally tolerated RASi
 - *watch out for K levels. If >5.5 → consider binder
 - May use steroidal MRA (aldosterone, eplerenone) if CHF, Conn’s, refractory HTN with caution in reduced GFR
- **GLP-1 agonist (Semaglutide) used in CKD-DM + gfr >25** + ACR >100 despite maximally tolerated RASi
 - Caution if Pancreatitis as cannot use, watch weight, liaise with endocrinologist, watch A1c

Sick Day Management

- Medications that may result in a decline in renal function when patient has a threat to their volume status OR may have decreased clearance
 - Instruct patients to hold these medications on days when they are feeling unwell, and re-initiate when they are feeling better/ volume status restored
- SADMANS medications
 - S- Sulfonylurea
 - A- ACEi
 - D- Diuretics
 - M- Metformin, MRAs
 - A- ARBs
 - N- NSAIDS
 - S- SGLT2i

MRA 101

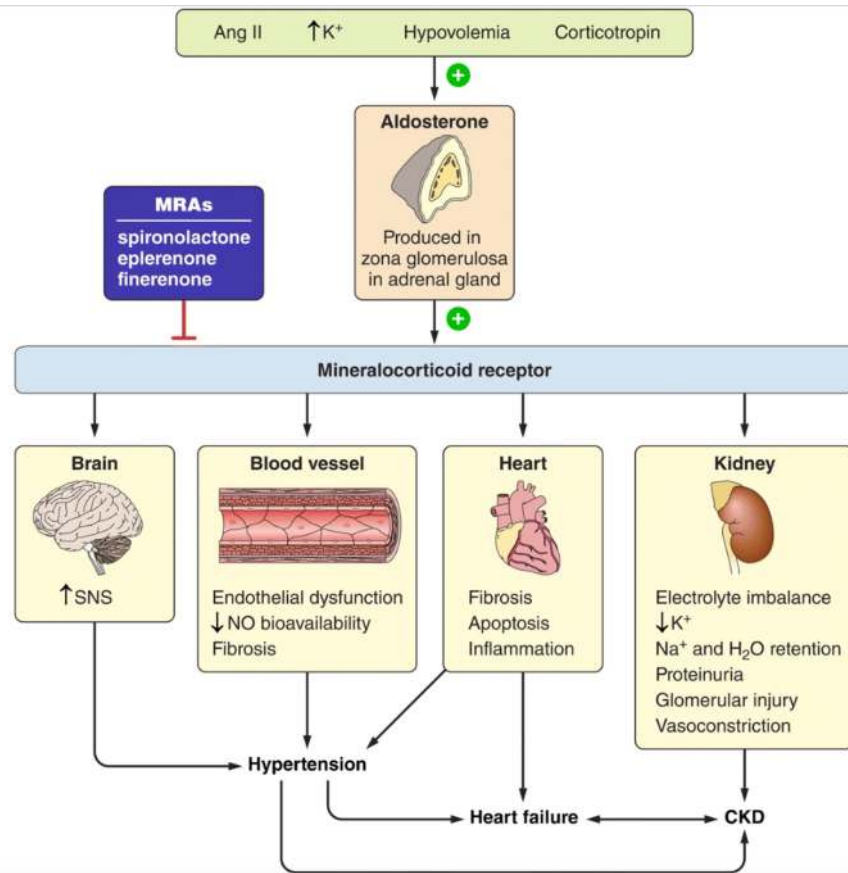


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1 Hbbs FDR et al *Nat Med* (2024). <https://doi.org/10.1038/s41591-024-03263-5> online

2 Honglei H et al *Int J Hypertens*. 2023 Mar 9;2023:6683987.

3 Pitt B et al *N Engl J Med* 2021;385:2252-2263.

4 Bakris GL et al *N Engl J Med* 2020;383:2219-2229

Effect	Spironolactone Steroidal	Eplerenone Steroidal	Finerenone Non steroidal
Potency to MR	+++	+	+++
Selectivity to MR	+	++	+++
CNS penetration	+	+	-
Sexual s/e (breast, gynecomastia)	++	+	-
Half life t1/2	>20h	4-6h	2-3 h
BP effect	↓↓↓	↓↓	↓ despite short t1/2
Effect on K^+ v placebo	↑↑	↑↑	↑
Effect on CKD	Worsening K , ↓ eGFR? ¹	Reduce ACR, BP but no benefit in eGFR, risk of hyper K ²	↓ progression CKD, ↓ CV and renal death, ↓ AFib (CKD w T2DM) ^{3,4}
Effect on CHF	↓↓↓ Less mortality any cause w HFrEF (RALES) ↓ HF hospitalization w HFpEF (TOPCAT)	↓↓↓ Less cardiac mortality and hospitalization w HFrEF (EMPHASIS-HF)	↓↓↓ HF hospitalization w HFpEF , non sig trend to ↓ mortality (FINEARTS) Study in HFrEF ongoing (FINALITY-HF) 100

8. CKD Complications



Management of Complications



Oral Exam Scenario

- Summary of complications / targets in CKD :
 - Hb 100-110, Tsat >30%
 - Hyperkalemia
 - Metabolic acidosis
 - PO₄ and Ca toward normal range
 - PTH target unknown for pre-dialysis CKD
 - PTH target for dialysis patients is 2-9x ULN
 - Hyperuricemia agents (for symptomatic gout)
 - Afib and CKD (NOAC preferred)

Management of Complications - hemoglobin

- Anemia: Rule out other causes of anemia, e.g. blood loss

If Hgb < 100:

- **1st step:** Iron studies

- Ferritin $\leq$ 500, Tsat $\leq$ 30%: Oral or IV iron (prefer IV if feasible & no Contraindications)
- Monoferric (ferric derisomaltose) or Venofer (iron sucrose)

- **2nd step:** Hb < 100 **and** *iron replete or after trial*

- Erythropoietin-stimulating agent (ESA)*: Eprex or Aranesp

- **Target Hb while on ESA: no higher than 115**

- Hb > 115 with ESA associated with stroke, CAD, HTN

*Caution in pts with: active cancer, previous stroke, uncontrolled BP

Management of Complications- K & H

- Hyperkalemia:
 - Potassium-restricted diet
 - Diuretics may help
 - K binders (sodium zirconium)
 - Avoid NSAIDs, look at other medications and diet
- Metabolic acidosis: start sodium bicarb tablets when bicarb <18

High K foods to watch out for:

- Fruits- **ORANGES**, tropical fruit (mango, papaya, pomegranate, kiwi, melons)
- Veg- **Avocado, tomatoes, potatoes**, beans, green leafy vegetables
- Other-Nuts/seeds, milk

K restricted diet = 1500 – 2700mg K/d

Beware Substitutes

"No Salt seasoning" = 650mg KCl / ¼ tsp

"Mio Water" (acesulfame potassium) = 30-90mg KCl / cup



Management of Complications – Ca & Phos

- Hyperphosphatemia: target normal PO_4
 - Decreased renal PO_4 excretion due to low GFR
 - Low PO_4 diet
 - Oral phosphate binders **with** meals
 - calcium-containing: e.g. calcium carb (TUMS) – **TYPICALLY FIRST LINE sec to cost**
 - But avoid if hypercalcemia, adynamic bone disease, vascular calcification
 - Non-calcium containing: e.g. sevelamer or lanthanum

High PO_4 foods to watch out for:

-Dairy- **CHEESE**, milk
Protein- shellfish, liver, **deli meats**

- Hypocalcemia: target normal Ca
 - Decreased 1- α -hydroxylase activity
 - Rx: calcium carbonate and calcitriol (1,25 Vit. D)
 - **Can't use vitamin D if hyperphosphatemic

Management of Complications - PTH

Target PTH 2-9x ULN in dialysis patients

- What if PTH is >9x ULN?
 - Calcitriol (1,25 Vit. D) but only if **PO₄** and **Ca** are not high
 - Cinacalcet (calcimimetic): activates Ca-sensing receptor to shut off PTH secretion
 - \$\$\$
 - Surgical parathyroidectomy in selected patients

Management of Complications - Osteoporosis

- CKD 1-2: with osteoporosis and/or high risk of fracture, manage as per general population
- CKD 3 +: **PTH in the normal range** and OP / high risk of fracture, manage as per general population (2B)
- CKD 3+: **with biochemical abnormalities of CKD-MBD** and OP or fragility fractures: “tx takes into account the magnitude and reversibility of the biochemical abnormalities and the progression of CKD, with consideration of a bone biopsy (2D)”...
 - In real life:
 - Can consider anti-OP medications if eGFR permits and optimized MBD parameters
 - **Most Bisphosphonates acceptable down to eGFR 30-35**
 - Beware of the risk of severe hypocalcemia with bisphosphonates or denosumab use in CKD
 - *Anti-OP meds are not the answer for dialysis patients*

Others

- Hyperuricemia + CKD
 - Consider uric acid lowering therapy in CKD patients after their first episode of gout (Xanthine oxidase inhibitors preferred)
 - For symptomatic acute gout in CKD – use low dose colchicine or steroids (NSAIDS less preferred)
 - Do not use for asymptomatic high uric acid levels
- Afib + CKD
 - Use oral anticoagulation (NOAC preferred*) for afib + CKD, medical therapy for hr <90, rhythm control if uncontrolled
 - *please renally adjust
 - *please refer / defer to your nephrologist regarding anticoagulation for afib in dialysis patient
 - Off label, not likely an MCQ therefore!

Indications for Dialysis

- **Urgent/In hospital indications: AEIOU:**
 - **A**cidosis (severe)
 - **E**lectrolyte problems, i.e. hyperkalemia
 - **I**ntoxication: methanol, ethylene glycol, Li, ASA
 - **O**verload, volume
 - **U**remia: pericarditis, encephalopathy/seizures
- **Dialyze if **REFRACTORY** to medical management**
- **Note: Anuria/oliguria is technically NOT in itself an indication to start**
- **Timing of dialysis in ICU
 - No clear benefit of early or late start. Individualize based upon patient factors (AKIKI trial Lancet 2021 and STARRT AKI (NEJM 2020). (see bonus slides for trial summaries)
- **Non-urgent/Outpatient**
 - Uremic symptoms/ Quality of life

BONUS

Read on own

Uremic Symptoms on History

- Anorexia, N/V, weight loss, appetite loss
- Metallic taste in mouth
- Pericarditis: typically doesn't usually show classic diffuse ST elevation
- Peripheral neuropathy
- Bleeding, i.e. platelet dysfunction
- CNS: mental status changes, seizures, mood disorder
- Amenorrhea, decreased libido, erectile dysfunction
- Nocturia, restless legs (related to iron deficiency)
- Pruritus

Bonus Read on
your own ☺

CKD in the CDA guidelines

- Note the different diagnostic level for albuminuria in diabetes

Stages of Diabetic Nephropathy by Level of Urinary Albumin Level

Stage of nephropathy	Urine dipstick for protein	Urine ACR (mg/mmol)	24-hour urine collection for albumin
Normal	Negative	<2	<30 mg/day
Microalbuminuria	Negative	2-20	30-300 mg/day
Overt nephropathy	Positive	>20 >67	>300 mg/day >1,000 mg/day

Values are for urinary albumin, not total urinary protein, which will be higher than urinary albumin levels. ACR results may be elevated with conditions other than diabetic nephropathy (see text and Table 4).

ACR, albumin to creatinine ratio.

Overview

1. ~~Acid Base~~
2. ~~Hypo/Hybernatriemia~~
3. ~~Hypo/Hyperkalemia~~
4. ~~Hypertension~~
5. ~~Glomerular Disease~~
6. ~~AKI~~
7. ~~CKD Progression~~
8. ~~CKD Complications~~

- BONUS Nephro Nuggets
 - Acronyms used
 - “Lumps” in the kidney**
 - ADPKD**
 - Choosing Wisely
 - Nephrolithiasis**
 - HTN (practical)
 - When to biopsy
 - Urine lytes/Hypokalemia (bonus)
 - Hematologic malignancies and the kidney**
 - Landmark trials
- Bonus MCQ Cases

Questions?

Lots of additional information and practice acid-base cases included in your slide deck.



Other causes of Proteinuria

Proteinuria Category	Level of Proteinuria	Examples
Transient	<1 g/day	Fever, heavy exercise
Orthostatic- proteinuria in upright position	<1-2 g/day	Pediatrics; uncommon after age 30
Overflow- overwhelmed resorption capacity of filtered protein	Variable (usually < 3.5g)	Myeloma, pigments (e.g. rhabdo, hemolysis)
Glomerular- increased filtration of protein	Variable; nephrotic range likely (>3.5g/day)	Diabetes, SLE, Amyloid, IgA, and primary causes
Tubulo-interstitial- impaired resorption of filtered protein	<2 g/day	Heavy metals, sarcoid, Sjogren's, NSAIDs (things that cause AIN!)
Post-renal proteinuria	<1g / day	Stones, genitourinary tumours, UTIs

Clinical evaluation of the kidney

- Urinalysis dipstick
 - Protein: hypertension, diabetes, MM, GN, orthostatic proteinuria, nonpathological (illness, fever, exercise, pregnancy)
 - Glucose: DM, pregnancy, SGLT 2 inhibitor, tubular dysfunction (MM)
 - Ketones: starvation, DKA, alcohol, poisoning
 - Leukocyte esterase and nitrites
- Urinalysis microscopy
 - Erythrocytes > 3
 - Isomorphic vs dysmorphic
 - + blood on dipstick with negative erythrocytes = rhabdomyolysis, hemolysis, mechanical valve
 - Leukocytes > 5 : infection
 - epithelial cells
 - Squamous: contamination. Renal tubular epithelial: ATN, transitional : ureters
- Casts:
 - Hyaline cast: CKD
 - WBC cast: AIN or infection
 - RBC cast: GN
 - Granular cast: ATN
 - Tamm Horsfall : normal
- Crystals
 - Calcium phosphate, calcium oxalate, uric acid, struvite

Imaging of the kidneys

- Ultrasound
 - Echogenicity: acute or chronic parenchymal disease
- Kidney stone: non contrast CT
- Hematuria: CT urography or cystoscopy
- Renal mass: MRI
- Doppler/CTA/MRA: renal artery stenosis

LUMPS IN THE KIDNEY

Renal Cysts - Bosniak classification (**based on CT**)

(CT preferred, however it is with Contrast! If CKD or contraindication to contrast: radiologists will also classify based on US)

- Bosniak 1 and 2: <5% are cancer
 - Simple cysts with thin walls, no (stage 1) or fine (stage 2) calcifications, and no (1) or thin (2) septae, no enhancement
 - **No need to follow up** 
- Bosniak 2F (F for 'follow-up') – approx 10% are cancer
 - **Smooth, minimally thickened wall. May have multiple septa, but smooth without contrast enhancement**
 - Needs active surveillance: repeat imaging at 6 mo, 12 mo, then yearly
- Bosniak 3: 50% are cancer
 - One or more enhancing **thick or irregular walls or septa**
- Bosniak 4: 90% are cancer
 - Like Bosniak 3 but also have a solid enhancing nodule

Bosniak 3 and 4 should be referred for partial nephrectomy

- If not fit for surgery, go for perc biopsy or active surveillance. Possible thermal ablation if candidate

LUMPS IN THE KIDNEY

Solid Renal Mass

- Image CT or MRI
- <1cm: too small to really workup
 - Needs active surveillance
- $\geq$ 1cm: vast majority are actually cancer
 - If scan shows lesion is clearly an angiomyolipoma, then no further w/u is needed
 - Otherwise, first check for metastases, then:
 - **If life expectancy >5 years, partial nephrectomy**
 - If life expectancy <5 years, or not fit for surgery, consider perc biopsy, active surveillance or thermal ablation (Beyond the scope of this exam!)

Renal Biopsy – once anathema due to fear of tumor seeding...

Whenever diagnosis is unclear (hematologic/ lymphoma vs. mets, inflammatory, infectious)

Required in Ontario prior to thermal ablation

ADPKD

- Polycystic Kidney Disease is an autosomal dominant disorder resulting in growth of renal cysts that destroy normal renal parenchyma
 - Affects 1/1000 live births
 - Common symptoms:
 - **HTN**
 - **pain- most common symptom; abdo pain common in cyst rupture**
 - hematuria- often from cyst rupture, proteinuria
 - Stones- **uric acid** stones , Ca Oxalate
 - UTI- 4 weeks antibiotics if infected cyst!
 - concentrating deficit- thirst, polyuria, polydipsia, nocturia
 - Erythrocytosis- ↑ epo levels
 - Extra-renal manifestations- liver cysts, diverticuli, **cerebral aneurysms**, pancreatic cysts
- Mitral Valve Prolapse/ AI most common cardiac anomalies**

ADPKD- 2025 Guidelines - Diagnosis

Initial Diagnosis can be made based on:

+Genetic testing (ADPKD1 or ADPKD2 gene) OR Family History, with LOWER cyst threshold (on U/S):

- ≥ 3 total if aged 15-39
- ≥ 2 /kidney if aged 40-59
- ≥ 4 /kidney if aged 60+

No Genetic testing OR Family History, with HIGHER threshold (needs to be based on MRI, not ultrasound):

- ≥ 10 cysts

Post-diagnostic ultrasound, **do follow up tests:**

MRI Kidney Volume (for Mayo Image Classification – not for GIM)

MRI Brain if neuro symptoms or FHx of IC aneurysm

ADPKD- 2025 Guidelines - Treatment

- Recommend all ADPKD patients be referred to nephrology for initial assessment, which should include imaging and consideration of genetic testing
- As on prior slide, **imaging**:
 - If known family history of PCKD– Ultrasound is serially preferred imaging
 - If no prior family history, US, MRI, or CT can be used -> Needs MRI eventually
 - Follow-up imaging should not exceed a frequency of once yearly
- **Treatment**
 - Sodium restriction to <2g/day, high fluid intake of 2-3 L/day
 - Start RASi to target BP<110/75 mmHg (if <50 y.o), but <120 mmHg (if >50 y.o)
 - Add Tolvaptan if (assuming eGFR>25):
 - eGFR <60 and declining 3+%/year, OR
 - Mayo Imaging Classification of 1C to 1E
 - Tolvaptan therapy side effects: Polydipsia, polyuria, hypernatremia, dehydration, liver enzyme elevation

Choosing Wisely Canada

- Don't start ESA in asymptomatic CKD pts with Hgb > 100 g/L
- Don't start NSAIDs in pts with HTN, HF, CKD
- Don't combine ACEi/ARB for HTN, diabetic nephropathy, CHF
- Don't start chronic dialysis w/o shared decision-making process between pts, families and health care team
- Don't start dialysis in CKD V without an indication

Nephrolithiasis

- Pathogenesis:
 - Increased excretion of solutes: calcium, oxalate, uric acid
 - Decreased urine volume
 - Abnormal urine pH
 - Alkaline: calcium phosphate, struvite
 - Acidic: calcium oxalate – most common, uric acid, cysteine (AR, genetic)
 - Absence of inhibitors: citrate, magnesium
 - Structural kidney disease
- General work-up (outside of acute setting):
 - Urinalysis, urine culture
 - Electrolytes, Cr, Ca, PO_4 , PTH, uric acid
 - 24 hour collection for volume, Na, Ca, PO_4 , uric acid, oxalate, citrate, Mg,
 - Urine Creat (to assess for adequate sample)

Heme Malignancies and the Kidney

- Myeloma Kidney (Seen with myeloma and other monoclonal gammopathies)
 - Cast nephropathy
 - Light chains both toxic to and clog tubules → AKI
 - Other mechanisms of AKI: hypercalcemia, hyperviscosity syndrome (Waldenstroms)
 - Tubular proteinuria, but dipstick negative (dipstick for albumin only!)
 - Fanconi's like syndrome
 - Risk factors: NSAIDs, Lasix, IV contrast
 - **Treat the myeloma**
 - Extracorporeal methods to remove SFLC (plasmapheresis and high cutoff dialysis) can be used as adjunctive therapy – *not great evidence and not for the scope of this exam*
- Amyloid Kidney
 - Deposition of fibrils in tissues
 - Congo red positive, apple-green birefringence
 - Deposit in glomerulus → leak albumin → nephrotic syndrome

AL amyloid: clonal light chains deposit in glomerulus

 - Treat with chemotherapy

Other: MGRS (monoclonal gammopathy of renal significance – multiple etiologies)

Note AA amyloid: chronic inflammatory disease leads to amyloid creation, deposits in kidney

 - Treat underlying disease: e.g. RA, TB, chronic osteomyelitis

BONUS CASES

IMR 2013-2024 participants have asked for “practice cases”.

Here are some of the cases we have presented in previous course years with explanations.

Hypo- and Hypernatremia- Case 1

A 45 year old female recently had a showering of strokes from infective endocarditis. A few days later you notice sodium trending up to 155. You initiate increased po intake of water but her sodium won't improve. Urine osm is 100. What is the next best step?

- a) Water deprivation test
- b) DDAVP trial
- c) ½ NS @ 45 cc/hour
- d) Supportive care

Hypo- and Hypernatremia- Case 1

A 45 year old female recently had a showering of strokes from infective endocarditis. A few days later you notice sodium trending up to 155. You initiate increased po intake of water but her sodium won't improve. Urine osm is 100. What is the next best step?

- a) Water deprivation test
- b) DDAVP trial**
- c) ½ NS @ 45 cc/hour
- d) Supportive care

-She has hypernatremia and low urine osm (dilute urine)
- This is diagnostic of DI
- She does not need a water deprivation test
- She has to have a DDAVP trial to differentiate between nephrogenic and central DI

Hypo- and Hypernatremia- Case 2

A 70-kg woman presents with symptomatic hyponatremia (seizure at home, confused in ED) ($\text{Na}^+ = 118$). You want to increase her sodium by 5 mmol/L using 3% saline. How much 3% NaCl should be given to achieve this?

- A. 250 mL
- B. 350 mL
- C. 500 mL
- D. 700 mL
- E. 1000 mL

Hypo- and Hyponatremia- Case 2

A 80-kg woman presents with acute symptomatic hyponatremia (seizure at home, confused in ED) ($\text{Na}^+ = 118$, last measured Na 139 one week ago). You want to increase her sodium by 5 mmol/L using 3% saline. How much 3% NaCl should be given to achieve this?

- A. 200 mL
- B. 300 mL
- C. 400 mL**
- D. 800 mL
- E. 1000 mL

$$\begin{aligned} \text{Correction: Volume infusate} &= (80\text{kg} \times 0.5 \times \text{planned rise Na}) \\ &= \frac{513}{200} \\ &= \text{Roughly } 400 \text{ ml} \end{aligned}$$

Acid-Base A Question

What is the acid base disturbance? pH 7.25, Na 132, cl 105, bicarbonate 12, albumin 20. Co2 20.

- a. Non anion gap metabolic acidosis
- b. High anion gap metabolic acidosis
- c. Mixed high anion gap and non anion gap
- d. Mixed high anion gap and non anion gap metabolic acidosis with respiratory alkalosis

Acid-Base A Answer

Answer is D: high anion gap and non anion gap metabolic acidosis with respiratory alkalosis

Ph 7.25, bicarb 12, co2 20 = a metabolic acidosis is the driving force

Anion gap is 15. But need to correct for albumin (for every decrease of 10 in albumin, add 2.5 to AG). Albumin decreased by 2x10, thus will add 2x2.5 to AG. $15 + 5 = 20$ is the true anion gap

Delta delta gap is $20 - 12 / 24 - 12 = 8 / 12 = 2 / 3 = 0.67 =$ high AG and NAGMA are both present

Check for the compensation, 1:1 ratio for metabolic acidosis

Hco3 changed by 12, c02 changed by 25 : 1:2 ratio, thus a metabolic alkalosis is also present

Acid-Base B Question

What is the acid base disturbance? pH 7.32, anion gap 20, albumin 40, bicarbonate 8, Co2 27.

- a. Metabolic acidosis with appropriate respiratory compensation
- b. Metabolic acidosis and respiratory alkalosis
- c. Respiratory alkalosis with metabolic compensation
- d. Non anion gap metabolic acidosis with a high anion gap metabolic acidosis

Acid-Base B Answer

Answer is D

Ph is 7.32, bicarbonate 8, CO_2 22 = metabolic acidosis is the driving force

Anion gap is 20. Albumin is 40 thus no need to correct

Delta delta gap is $20 - 12 / 24 - 8 = 8 / 16 = 0.5$ = the bicarb changed much more than expected – thus a NAGMA and HAGMA is present

Check the compensation Metabolic acidosis has a 1: 1 change

Bicarb changed by 16, CO_2 changed by 18, thus a 1:1 ratio. Thus CO_2 has appropriately changed.

Acid-Base C Question

What is the acid base disturbance? pH 7.12. Bicarbonate 20. Na 141, Cl 105, K 2.5. Albumin 38. CO₂ 40.

- a. Metabolic acidosis with appropriate respiratory compensation
- b. Metabolic acidosis and respiratory alkalosis
- c. Respiratory alkalosis with metabolic compensation
- d. Non anion gap metabolic acidosis with a high anion gap metabolic acidosis

Acid-Base C Answer

Answer is A

Ph 7.12, CO_2 20, HCO_3^- 12 = metabolic acidosis is the driving force

Anion gap is $141 - 105 - 20 = 16$.

Albumin is 38 thus no change needed.

Delta delta gap is $16 - 12 / 24 - 20 = 4 / 4 = 1$ = only a HAGMA is present

Compensation is 4:5, roughly a 1:1 ratio – thus only a HAGMA with appropriate respiratory compensation is present

Acid-Base D Question

A woman is found unresponsive, surrounded by empty pill containers, and bottles.

On exam: BP 120/70, HR 105, T 37.2, RR 28

Labs: Na 136, K 4.2, Cl 100, HCO₃ 16, BUN 4, Glc 6, Posm 290, pH 7.58, pCO₂ 25. Alb 40

What was the most likely cause of her acid-base disturbance(s)?

- A)Methanol
- B)Oxycodone
- C)Lorazepam
- D)ASA

Acid-Base D Answer

Answer is D) ASA

- pH is high and PCO₂ is low → **respiratory alkalosis** → *unlikely opioid / benzo intoxication* (would expect resp acidosis)
- Delta CO₂ = 40 – 25 = 15
- Delta HCO₃ = 24 – 16 = 8 → HCO₃ drops more than expected, likely concurrent **metabolic acidosis**
 - In acute resp alk: For every drop in CO₂ by 10, HCO₃ should drop by 2.
 - Expected HCO₃ drop is 3 (for a drop in CO₂ of 15). *Actual delta HCO₃ = 8 (drop is higher than expected, therefore something driving bicarb down)*
- AG = 136 – 100 – 16 = 20 → elevated **AGMA**
- Delta- delta: delta AG = 8, delta HCO₃ = 8, *therefore pure AGMA*
- Osmolar gap = 8 (normal) → *less likely toxic alcohol ingestion (unless all metabolized, however more likely ASA OD)*

Resp alk and AGMA: Most consistent with ASA overdose

Acid-Base E Question

A 23 year old male presents with weight loss. His blood pressure is 101/65, Hr 99, T 36.6, oxygen 99%. His bloodwork shows Na 135, K 2.3, bicarbonate 9, Cl 118. Urinary studies show Na 20, K 5, Cl 29.

- a. Gittleman syndrome
- b. Laxative abuse
- c. Diabetes
- d. RTA 2

Acid-Base E Answer

Answer: B. This person has a non anion gap metabolic acidosis.

Urinary gap is negative suggesting a GI pathology (ie laxative abuse). This also explains the low potassium, low pressures and mildly high heart rate, as well as case presentation of weight loss.

Answers c and D are both RTA's, but would have a positive urine anion gap.

Also, Gittleman syndrome presents with alkalosis.

Acid-Base F Question

A 58 year old male presents with loss of consciousness. His vitals are 100/60, hr 98, T 37, oxygen 99%. His blood work shows Na 142, K 2.4, bicarbonate 32, Cl 102, glucose 3. Urine electrolytes show urine Na 12, urine K 5, urine cl 19. ABG shows a pH of 7.40, CO2 43. What is the likely etiology?

- a. Vomiting
- b. Bartter's syndrome
- c. RTA 4
- d. Primary hyperaldosteronism

Acid-Base F Answer

Answer: A (vomiting)

This person has a metabolic alkalosis with a low urine chloride and low blood pressure. This is suggestive of vomiting.

Bartter's syndrome presents with high urine chloride. RTA is a metabolic acidosis.

Primary hyperaldosteronism would have hypertension.

Acid-Base G Question

A 44 year old male with weight loss and abdominal pain presents with the following bloodwork: Na 141, Cl 113, K 2.2, bicarbonate 18, glucose 8. Urine pH is 5.2, Urine Na 20, Urine K 10, Urine Cl 15. What is a possible underlying pathophysiology?

- a. Multiple myeloma
- b. Diabetes
- c. Urinary retention
- d. Kidney stone

Acid-Base G Answer

Answer is A (multiple myeloma)

He has a non anion gap metabolic acidosis with a positive urine gap suggestive of RTA. Because of hypokalemia he has RTA 1 or 2.

Because urine pH is 5.2, this suggests more RTA 2, associated with multiple myeloma. Diabetes, urinary retention are associated with RTA 4
Calcium phosphate stones are associated with RTA type 1.

Acid-Base H Question

A 25 year old male presents with decreased level of consciousness. His vitals are within normal limits. His bloodwork shows Na 141, urea 3, K 4.4, glucose 5, Cl 101, Hco3 8, serum osmolarity 318. His ABG shows a Ph 7.22, Co2 22.

- a. Fomepizole
- b. Dialysis
- c. IV bicarbonate
- d. IV normal saline

Acid-Base H Answer

Answer is A:

Anion gap = $141 - 101 - 7 = 34$

Serum osm = $2 \times 141 + 5 + 3 = 290$

Osm gap = $318 - 290 = 28$ (elevated)

He has an anion gap metabolic acidosis with an osmolar gap suggestive of ethylene glycol poisoning.

A Co_2 of 22 is appropriate compensation for a bicarbonate of 8.

Acid-Base I Question

A 55 year old male presents with altered mental status and fruity smelling breath. His vitals are stable. His bloodwork shows Na 134, Cl 115, K 4.5, Hco₃ 12, glucose 2, urea 8, serum osmolarity 302. What is your first step in management?

- a. IV normal saline
- b. IV ethanol
- c. IV dextrose
- d. Dialysis

ACID-BASE I Answer

Answer is C

She has a non anion gap metabolic acidosis, with an elevated osmolar gap suggestive of Isopropyl alcohol poisoning.

Treatment is supportive.

Thus manage his hypoglycemia with D10W.

IV normal saline would be needed if patient was hypotensive etc

IV ethanol is needed in ethylene glycol poisoning.

Patient does not have any indications for dialysis at this point.

ACID-BASE J Question

35 year old female presents with fatigue. She has no past medical history and is not on any medications. Her vitals are hr 92, bp 110/50, T 37, oxygen 99%. Her bloodwork shows Na 136, K 2.9, Cl 95, Hco3 31. Her urine electrolytes show Urine Na 60 mmol/L, urine chloride 35. What is the etiology of her findings?

- a. Conn syndrome
- b. RTA type 4
- c. Gittleman Syndrome
- d. Villous adenoma

Acid Base J Answer

Answer is C

She has hypokalemic metabolic alkalosis (not b).

Conn syndrome is hyperaldosteronism and will present with hypertension.

Gittleman syndrome presents with normotensive or hypotensive metabolic alkalosis with a high urine chloride (>20).

Villous adenoma presents with acidosis.

ACID-BASE K Q & Answer

You are seeing a 32M for follow up in your clinic. He has a PMHx of HIV and is on ARVS (Abacavir, tenofovir, and dolutegravir). He has no other past medical history, and no current complaints. His labs are as follows:

Na- 135 K-3.2 HCO₃⁻- 20 Cl-105 PO₄⁻- 0.7 Cr- 115

Urinalysis- no blood, + glucose, trace protein

Urine Lytes- — U_{Na}-34, U_K-25, U_{Cl}-28

What is the cause of his lab abnormalities?

- A. Laxative abuse
- B. Type IV RTA
- C. Sjogren's disease
- D. Fanconi syndrome

Answer D) Secondary Fanconi's is a known side effect of tenofovir therapy! It is difficult based off symptoms or just the acid/K level to truly tell type 1 and type 2 apart. LOOK FOR CLUES- mention of stones, low PO₄, glucosuria UAG = positive, a renal cause!

Electrolytes Bonus 1 Q&A

A 65 year old alcoholic presents with a sodium of 122. His weight is 60 kg. He has no neurological deficits or changes. You are unsure if this is chronic or acute. What fluids will you run if you would like to safely correct his sodium?

- a. IV dextrose
- b. Normal saline at 58 ml/HR**
- c. Half normal saline at 90 ml/HR
- d. Hypertonic saline bolus

TBW = $60 \times 60\% = 36\text{kg}$.
Desired sodium change = 6
 $36 \times 6 / 154 = 1.4\text{L}$ over 24 hours = 58 ml/hour of normal saline
Hypertonic bolus is not needed

Electrolyte Bonus 2 Q&A

A 65 year old female in the ICU has a sodium of 175. Her weight is 60 kg. What fluids would you run if you would like to safely correct her sodium by 10?

- a. IV dextrose at 75 ml/hour
- b. Half normal saline at 100 ml/hour
- c. Normal saline at 75 ml/hour
- d. IV bicarbonate at 23 ml/hour

Answer A
(Serum Na-desired Na)/desired Na x weight x %water = water deficit
 $175-165/165 \times 60 \text{ kg} \times 0.5\% = 1.8 \text{ L in 24 hours}$
 $1.8/24 = 75 \text{ ml/hour}$

Electrolyte Bonus 3 Question & Answer

A 65 year old male presents with abdominal pain and weight loss. His blood work shows Na 141, K 6.9, bicarbonate 24, creatinine 331, phosphate 4, calcium 2.0, magnesium 2.2, uric acid 696. What is the most likely cause of his AKI ?

- a. Ethylene glycol poisoning
- b. Sickle cell disease
- c. Urinary obstruction
- d. Tumor lysis syndrome

Answer D) Tumor lysis syndrome will present with AKI, hyperkalemia, hyperphosphatemia, uricemia, hypocalcemia.
Ethylene glycol poisoning will present with elevated anion gap and serum osm gap
Urinary obstruction will not cause hyperphosphatemia or hypocalcemia

Electrolyte Bonus 4 Question & Answer

A 23 year old female (weight 55 kg) presents with new onset seizures. Her bloodwork shows Na 102, K 4.5, Cr 65, bicarb 23. What is your next step in management?

- a. 270 ml bolus of hypertonic saline
- b. IV normal saline at 53 ml/min
- c. IV half normal saline at 48 ml/min
- d. IV magnesium

Answer (a) – 270 ml bolus hypertonic saline
Patient requires hypertonic saline due to seizures from hyponatremia
$$TBW \times 5 / Na \text{ infusate}$$
$$55 \text{ kg} \times 50\% \times 5 / 513 = 268 \text{ ml of hypertonic saline}$$

Electrolyte Bonus 5 Question & Answer

A 33 year old male presents with nausea, vomiting and fatigue. He has a history of bipolar disorder on lithium. His blood work shows Na 149, K 4.5, creatinine 65, calcium 3.1. Serum osm 303, urine osm 102. What is the next step and most likely underlying pathophysiology

- a. Water deprivation test; Psychogenic polydipsia
- b. Monitor response to DDAVP; Nephrogenic diabetes insipidus
- c. CT head; Central diabetes insipidus
- d. Water deprivation test; Dehydration

Answer (b) – Monitor response to DDAVP;
Nephrogenic DI. Most likely in the setting of lithium use. No need to do water deprivation test if hypernatremic and low urine osm.

Electrolyte Bonus 6 - answer

You are seeing a 25 year old female in your office who has been referred by her GP for hypertension. She otherwise is healthy and has no past medical history. Her only medication is OCP. She has a family history of HTN in her mother, maternal aunt and maternal cousin. Lab work shows:

Labs: Na- 137 K-3.1 HCO₃-29 Cr-67; low renin/ low aldo levels, AM cortisol-448, TSH-3.6, bHCG negative

Urine lytes: U_K- 40 U_{Na}- 50 U_{Cl}- 35

What is the diagnosis?

- A. Liddle's syndrome
- B. Conn's syndrome
- C. Gitelman's
- D. Villous Adenoma

Answer (a) – Liddle's *Liddle's is ENAC channel upregulation in distal nephron → Htn/ hypok with low renin & low aldo

Hypokalemia case:

-Urine K =40 therefore renal K loss → unlikely villous adenoma
Step 2: check BP :Hypertensive → Liddle's or Conn's
Step 3: check renin/ aldo

-Low renin / low aldo: Liddle's >> Conn's (expect high aldo)
Unlikely Gitelman: expect normo/hypotension

Unlikely villous adenoma: expect low urine K and negative UAG

Electrolyte Bonus 7

A 20 year old female presents to your office with the following labs. She was recently prescribed aminoglycosides. Na 135, K 2.0, Cl 110, Mg 0.5, Hco₃ 29. Urine studies show an elevated urine potassium, sodium, calcium and chloride. Her vitals are 95/68 with a HR of 78. What is her likely diagnosis?

- a) Liddle's syndrome
- b) Barters syndrome
- c) RTA type 2
- d) Reninoma

Electrolyte Bonus 7- Answer

A 20 year old female presents to your office with the following labs. She was recently prescribed aminoglycosides. Na 135, K 2.0, Cl 110, Mg 0.5, Hco3 29. Urine studies show an elevated urine potassium, sodium, calcium and chloride. Her vitals are 95/68 with a HR of 78. What is her likely diagnosis?

- a) Liddle's syndrome
- b) Barters syndrome**
- c) RTA type 2
- d) Reninoma

Bartter's syndrome (mimics Lasix)
Will present with
- low K
- alkalosis
- low normal BP
- Urine lytes (esp calcium!) will be elevated

- RTA type 2 would have a low bicarb (acidosis)
- Reninoma and Liddle's would have hypertension

GN Bonus Case 1 Q&A

Which of the following is a relative contraindication of renal biopsy?

- a. Solitary kidney
- b. Pregnancy
- c. Contrast allergy
- d. Hematuria

Answer (a) Solitary kidney is a relative contraindication
Pregnancy is not (you can position the patient accordingly)
Contrast is not used during kidney biopsy
Hematuria and proteinuria are indications TO DO a biopsy

GN Bonus Case 2 Q&A

A 68 gentleman presents with chronic dyspnea and wheezing. In the last few weeks, he's experienced some weight loss and weakness.

No hemoptysis, or clear infectious signs or symptoms. No meds.

He looks frail, but hemodynamics are stable without O2 requirements, and afebrile.

Labs are notable for a Serum creatinine of 180 micromol/ L, (Baseline 90 12 mos ago). Electrolytes and CBC WNL. UA: blood and protein.

CXR is unremarkable. Urine microscopy shows dysmorphic RBCs and red cell casts.

You send a work up, and request a stat ANCA which comes back anti PR3 positive. The rest of the GN work up is pending.

What would be the best therapy or next step:

- A) 50 mg Prednisone + plan for Cyclophosphamide
- B) Renal biopsy
- C) PLEX
- D) 1 g Methylprednisone + plan for Rituximab

Answer (D) 1g Methylprednisolone + plan for Rituximab

GN Bonus Case 3 Q&A

A 32 year old female with no past medical history presents with R flank pain, fever, and hematuria. She trialled amoxicillin and NSAID therapy with no relief. Her work up shows normal bloodwork except for a creatinine of 150. In the office she has mild hypertension. Her urinalysis shows blood and protein. Her renal ultrasound does not show hydronephrosis. Her CT scan of the pelvis is clear. What is the underlying etiology

- a. NSAID induced AKI
- b. Nephrolithiasis
- c. IgA nephropathy
- d. post-infection GN

Answer (c) IgA nephropathy presents with nephrolithiasis but with a negative CT scan, as well as mild proteinuria with hematuria.

GN Bonus Case 4 Q&A

A 28 year old drug user presents with palpable purpura, abdominal pain, acute kidney injury and digital ischemia. What is your treatment

- a. Sofosbuvir alone
- b. Prednisone alone
- c. Prednisone, rituximab and sofosbuvir

Answer (C) prednisone, rituximab and sofosbuvir.
patient has cryoglobulinemic vasculitis of moderate to severe intensity requiring steroids with rituximab

GN Bonus Case 5 – Q&A

A 43 year old female with a history of membranous nephropathy presents with bilateral flank pain and hematuria. Her GN was well controlled with RAAS blockade until this episode. Her most recent bloodwork shows creatinine of 165, albumin 13, and urinalysis shows 2+ proteinuria. What is the management for the most likely etiology?

- a. Increase candesartan
- b. Dialysis
- c. Anticoagulation
- d. Pulse steroids

Answer (c)
This person with low albumin, bilateral flank pain, hematuria in the setting of membranous nephropathy likely has renal vein thrombosis.

GN Bonus Case 6 - Question

A 58 year old female has routine blood work showing Na 138, K 4.5, Cr 213, bicarb 19. Urinalysis and renal ultrasound are unremarkable. She is completely asymptomatic with no physical exam abnormalities. She has a history of hypertension, well controlled on amlodipine. Her most recent blood work 2 weeks ago showed normal renal function. At that time she had otitis and was started on amoxicillin. What is the most likely etiology of her AKI

- a. Acute interstitial nephritis
- b. Post-infectious GN
- c. Papillary necrosis
- d. Hypertensive nephropathy

Answer (a)
Most likely has AIN in the setting of amoxicillin. Unlikely to be post-infectious GN without hematuria or proteinuria. Unlikely to have hypertensive nephropathy with well controlled blood pressure and sudden rise of creatinine.

GN Bonus Case 7

A 35 y M presents with R sided flank pain for the past 3 days. He notes he has been seeing blood in his urine as well. Movement worsens the pain, and there are no aggravating factors. He has been taking Ibuprofen 400 TID x3days for the pain. He denies fever, chills but endorses nausea and vomiting. He has no prior medical history and is not on medications. In the ER his vitals show bp is 143/92, hr 102, O2 99%, T 36.8. His labs show creatinine 164 (baseline not known), K 3.8, bicarb 24. UA shows 1 + hematuria, 1 + proteinuria, 2 WBC. CT scan is unremarkable. Microscopy is unremarkable.

What is the management for the most likely diagnosis next step?

- a. Pulse steroids
- b. Renal biopsy
- c. Obtain 24 hour protein collection and start ACE/ARB when creatinine stabilizes
- d. IV fluids, pain control, straining urine, urology referral for possible stent placement for nephrolithiasis
- e. Abdominal ultrasound with doppler

Answer: © This is likely IgA nephropathy, which can present similar to nephrolithiasis but CT scan will be negative. You want to quantify 24 hour protein and start ACE/ARB when safe to do so. Steroids are needed if patient has uncontrolled proteinuria after 3 months of management. Biopsy is the only way to confirm diagnosis of IgA, but is rarely used up front as it does not impact management.

Hypertension Bonus Case 1

A 28-year-old woman with HTN is referred for assessment. She has no major comorbidities. Exam reveals a BMI of 20 and an abdominal bruit.

She is currently on amlodipine, at maximum dose. Renal ultrasound performed last week showed a right kidney of 10.2 cm and left kidney 8.4cm. Both kidneys had normal echogenicity and no hydronephrosis.

Her BP in your office by AOBP is 155/100. HBPM 150/95. Basic electrolytes and Creatinine are normal

Based on the latest Hypertension Canada guidelines, what do you recommend to screen for her suspected condition?

- a) Urine metanephrines and catecholamines
- b) Order a captopril renal scan
- c) Order renin and Aldosterone level
- d) Order an MRA abdomen

D) Order MRA Abdomen

Hypertension Bonus Case 2

A 43 year old female with newly diagnosed hypertension is found to have a R sided adrenal mass. What is your next step?

- a. Adrenal mass biopsy
- b. Adrenal vein sampling
- c. Removal of R sided adrenal mass
- d. Aldosterone therapy
- e. Aldosterone:Renin Ratio

E) Aldosterone : renin ratio

Hypertension Bonus Question 3

A 68 year-old female has multiple blood pressure readings of 135/85, no orthostatic drop. Her work-up reveals creatinine of 123 $\mu\text{mol/L}$ (egfr 40 ml/min), A1c 4.5%, LDL 1.8, ACR 30 mg/mmol, K 4.5. EKG NSR, no LVH. She is currently not on any medications. What are your recommendations?

- a) Nothing, her blood pressure control is acceptable
- b) Start Amlodipine with a goal SBP of < 130 SBP
- c) Start Hydrochlorothiazide with a goal SBP of < 140
- d) Start candesartan with a goal SBP < 120

Answer, (d) Explained: Target is <120 as this person is > 50 YO + CKD, Sprint trial candidate, D is also preferred as want ARB w her proteinuria.

Hypertension Bonus Case 4

Which of the following patients would you screen for renovascular hypertension?

- a. A 43 year old female alcoholic with a bp of 145/90
- b. A 25 year old male with newly diagnosed hypertension and worsening creatinine by 30% after initiation of lisinopril
- c. A 55 year old male with blood pressure controlled on maximum doses of lisinopril, amlodipine and hydralazine.
- d. A 48 year old female with new onset pulmonary edema and bp of 165/80.

Case 4: Answer: B – age < 30 and creatinine rise with ACE. Patient in C has controlled blood pressure on 3 agents

AKI Bonus Question 1

A 70 year-old man has an ACS. ASA, Plavix and pantoloc are started.

1 month later, presents to GP with an upper respiratory tract infection. Creatinine has risen from 80 to 150.

PMHx: CAD, DM2, HTN, ex-smoker

HPI: mild peripheral edema for 2 weeks

U/A: 1+ protein, 1+ RBCs, 1+WBC, +WBC casts

CBC: mild peripheral eosinophilia

Most likely cause of acute kidney injury?

- a) NSAID-induced minimal change disease
- b) AIN
- c) Pyelonephritis
- d) Lupus nephritis

Answer B. Note that urine eosinophils have low SN and SP for AIN – but exams refer to it anyway, ask for Hansel stain
AIN 2ry to pantoloc / PPI
NSAID-induced minimal change will not lead to eosinophilia

CKD question 1- 2025

50 year old male with gfr of 30 ml/min. BP is 115/75, K 4.5, LDL 2.
What changes would you make today?

1. Increase antihypertensive
2. Start K binder
3. Start statin
4. Nothing

CKD question 2 2025

- 45 year old male with CKD stage 3. BP is 120/70, K is 4.5, bicarb is 17, proteinuria is 600 mg/day. He is not yet on any medications. What would you do?
 1. Start RASi + sodium bicarbonate
 2. Start RASi
 3. Start sodium bicarbonate
 4. Do nothing

CKD question 3 2025

- Who would be a candidate for starting SGLT2i
 1. DM + gfr 15ml/min
 2. CHF + gfr 19 ml/min
 3. No comorbidities and gfr 30 ml/min
 4. Everyone

CKD question 4 2025

- 65 year old male with CKD 5 comes in very tired. His hemoglobin is 92. His iron saturation is 10% and ferritin 25. What is your management regarding his anemia?
 1. Give iron (IV or PO depending on availability and preference)
 2. Start EPO
 3. Start dialysis
 4. Do nothing

CKD Question 5 - 2025

A 68yM with T2DM and CKD eGFR 25 ml/min/1.73m² presents for regular review. Meds include candesartan 32mg daily, rosuvastatin 10mg daily, empagliflozin 10mg daily, linagliptin 5 mg daily, glargine 24u daily.

BP 124/78. BMI 28.2

Na 140 K4.9 HCO₃ 22 LDL 1.5.

Hgb 106 ferritin 180 Ca 2.17 alb 38 Phos A1C 6.4%

Review of his CGM reveals Time in Range 92% (glucose 3.9 – 10mmol)

ACR 9.2 mg/mmol

Which of the following is next best choice for his CKD optimization:

- 1) Discontinue linagliptin and start semaglutide
- 2) Discontinue candesartan and start perindopril 16 mg /d
- 3) Add finerenone 10 mg daily
- 4) Add indapamide 1.25 mg daily

CKD Question 6 - 2025

A euvolemic 49 year old diabetic male with history of CKD 3 presents for routine follow up and the following labs. His medications include empagliflozin 10 mg daily, perindopril 8 mg daily, metformin 1000 BID. Na 141, K 5.4, bicarb 24. Along with recommending a low-potassium diet, what are your recommendations?

1. Stop perindopril
2. Start potassium binder
3. Start Lasix 40 mg mg daily
4. Start calcium supplementation daily

CKD Questions 1-6 Answered

Question 1 answer: 3. start statin

(if SGLT2i was listed that would be an appropriate answer too)

Question 2 answer: 1. start RASi (until maximally tolerated for the proteinuria) and sodium bicarb for acidosis

Question 3: answer 3: SGLT2i for CKD with gfr 20-45

Question 4: answer 1: start iron. Once iron replete can give epo.

Question 5: Answer 3 : start finerenone. He has albuminuria above target despite max dose RASi. Best to reduce risk of progression of CKD. GLP1RA is recommended to add for optimization if unable to achieve glycemic targets and he has tight control already.

Question 6: CKD question – answer 2, potassium binder.

Sample Oral Scenario:

A 67F with ischemic nephropathy (likely due to hypertension) is seen follow-up. She complains of fatigue and shortness of breath. She also has a history of CAD, DLP, and impaired glucose tolerance. She is a smoker, 10 cigarettes per day, 20pk yr history.

Medications: calcium carbonate 500 mg BID with meals, calcitriol 0.25mcg daily, pantoprazole 40 mg daily, amlodipine 10mg daily, replavite 1 PO daily, atorvastatin 40mg daily.

BP 165/94, HR 72, Afebrile. 2+ pretibial edema, lungs bibasilar crackles, O2 95%.

Labs: Hgb 95, ferritin 341, Tsat 15%

Cr 600 (last 535 1 month ago), K 5.5, HCO₃ 18, PO₄ 1.2, Ca 2.1, PTH 21, Albumin 34. Glucose 6.8, Hb A1C 6.3%. Urinalysis 1+ proteinuria, neg blood. Proteinuria < 1 g / day last quantified

Please outline your approach to management of this patient.

Sample Oral Scenario

Approach to the CKD consult:

- Acid/base
- Anemia
- BP
- Electrolytes
- Fluid balance
- MBD [Metabolic bone disease]
- Uremia
- Diabetes
- CVD risk factors
(≥ 50 yrs and $GFR < 60$ = statin if not on dialysis!)
- Vaccines- Flu shot!
- “Sick day management”
- Modality/Access

CKD Scenario

- For our patient:
 - **Acid/base**- start bicarb tablets for <18
 - **Anemia**- Hgb below target: start Iron supplementation, then darbopoetin 0.45mcg/kg SC weekly (often done concomitantly however try to restore iron stores first!)
 - **BP**- above target, additional antihypertensive- diuretic given fluid overload , **probably a sprint CANDIDATE so assuming standing BP is >110 can target SBP <120**
 - **Electrolytes**- dietary counseling for hyperkalemia:
 - **Volume status**: Lasix to help manage fluid and hyperkalemia
 - **MBD**- Ca and PO4 on target with calcitriol and CaCO3, PTH target unknown in CKD; can consider DEXA
 - **Uremia**-no significant uremic symptoms
 - **Diabetes**- dietary and exercise for pre diabetes
 - **CVD risk factors**- On statin
 - **Vaccines**- Covid vaccine, flu shot
 - **Other issues**- smoking cessation
 - SGLT2 I for CKD +/- diabetes

2026: Bonus Q 1 - HTN

A 52-year-old man presents with newly diagnosed hypertension. His blood pressure today is 148/85 mm Hg. Laboratory testing shows an eGFR of 63 mL/min/1.73 m² and a serum potassium of 4.6 mmol/L. He has no history of diabetes, cardiovascular disease, or albuminuria. What is the most appropriate initial management step?

- A. Start an ACE inhibitor
- B. Start an ARB or thiazide diuretic
- C. Advise a low-salt diet
- D. Start a calcium channel blocker

B: Start an ARB or TZD Diuretic. HTN Canada 2025 suggests combo therapy as start if SBP >= 140 mmHg or >= 130 mmHg with inc CVD risk.

2025: Bonus Q 2 - HTN

A 36-year-old woman is evaluated for hypertension and persistent hypokalemia. Her blood pressure is 155/92 mm Hg. She takes a combined oral contraceptive pill (ethinyl estradiol and drospirenone) and no other medications. Laboratory testing shows:

- Serum potassium: 3.2 mmol/L
- Plasma aldosterone concentration: 300 pmol/L
- Direct renin concentration (DRC): 100 pmol/L
- Aldosterone-to-renin ratio (ARR): 3

What is the most appropriate next step in management?

- A. Start spironolactone
- B. Send for adrenal vein sampling
- C. **Stop the oral contraceptive pill**
- D. Start an ACE inhibitor

C) You MUST REPEAT ARR off OCP as this causes false negatives. Obviously in this young person also need to ensure avoid pregnancy during workup of condition.

2025: Bonus Q 3 - GN

A 42-year-old Caucasian man with biopsy-proven IgA nephropathy is being followed in clinic. His biopsy shows a MEST-C score of M1 E1 S0 T0 C0. He is currently treated with an angiotensin receptor blocker (ARB) and an SGLT2 inhibitor. Despite optimal supportive therapy, his proteinuria remains 1.4 g/day, and his eGFR is 68 mL/min/1.73 m².

What is the most appropriate next step in management?

- A. Observation
- B. Start mycophenolate mofetil (MMF)
- C. Start hydroxychloroquine
- D. Start prednisone at 1 mg/kg/day
- E. Start prednisone at 0.5 mg/kg/day

E) 2025 GN IgA nephropathy guidelines suggest pred 0.5 mg/kg IF >0.5 g/day proteinuria.

2025: Bonus Q 4 - CKD

A 59-year-old man with type 2 diabetes mellitus and diabetic kidney disease is being followed in clinic. His current medications include losartan, empagliflozin, and finerenone. His blood pressure is 128/78 mm Hg. Laboratory studies show:

- eGFR: 38 mL/min/1.73 m²
- Urine albumin-to-creatinine ratio (UACR): 470 mg/mmol
- HbA1c: 7.8%

Despite optimized therapy, his proteinuria persists.

What is the most appropriate next step in management?

- A. Increase finerenone dose
- B. Add a GLP-1 receptor agonist
- C. Switch from SGLT2 inhibitor to DPP-4 inhibitor
- D. Start low-dose prednisone
- E. Add a thiazide diuretic

B) 2024 CKD KDIGO Guidelines suggest GLP1 agonists for patients with diabetic kidney disease and eGFR 25-75 to reduce proteinuria & CKD progression.

2025: Bonus Q 5 - CKD

A 64-year-old man with hypertension and CKD (eGFR 52 mL/min/1.73 m²) is being evaluated in clinic. His baseline blood pressure is 148/88 mm Hg and UACR is 35 mg/mmol. He is started on an ACE inhibitor.

Two weeks later:

- BP: 118/76 mm Hg
- Serum potassium: 5.4 mmol/L
- eGFR: 50 mL/min/1.73 m²

What is the most appropriate next step in management?

- A. Stop the RAS inhibitor
- B. Start an SGLT2 inhibitor**
- C. Start a thiazide diuretic
- D. Add loperamide (sodium zirconium cyclosilicate)

B) Given BP controlled, no indication for thiazide.
Given hyperk mild, do NOT need to stop RASi or start K binders.
SGLT2i helps proteinuria AND lowers K.

2025: Bonus Q 6 - GN

A 58-year-old asymptomatic woman presents with edema, albuminuria (4 g/day), and low albumin (28). She is diagnosed with membranous nephropathy on kidney biopsy. She is otherwise asymptomatic.

She is otherwise healthy. She is a non-smoker. Her routine preventive care includes: last mammogram 3 years ago, FOBT 1 year ago, and Pap smear 2 years ago.

Given her current presentation, what is the next best test to perform?

- A. Low-dose CT of the chest
- B. Colonoscopy
- C. Mammogram
- D. Anti-dsDNA

(C) Membranous nephropathy patients should undergo ROUTINE cancer screening, NOT extra testing (i.e. colonoscopy or CT chest. Mammogram should be q2years so is out of date. Anti DS DNA is reasonable but is not the NEXT test to perform. First rule out cancer then workup other causes of membranous nephropathy.

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